Diffpurifier: An Optical and SAR Image Change Detection Method Based on Diffusion Purification

Yiquan Xu, Xin Su, *Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Disasters occur continuously around the world, causing significant damage to human life and property. In practice, there are inevitably situations that require the use of heterogeneous remote sensing images, for instance, optical and synthetic aperture radar (SAR) images, for change detection (CD) and disaster recognition. However, the significant differences between heterogeneous remote sensing images make direct comparison difficult. Although some methods use image translation (IT) to reduce these differences, most employ a two-stage "translation followed by CD" strategy, which can lead to feature degradation. Furthermore, these methods often require adversarial training or complex loss functions that are sensitive to hyperparameters. This article proposes a CD network for optical and SAR images, named Diffpurifier. First, optical images are translated into SAR images using pre-trained denoising diffusion probabilistic models (DDPMs) and ordinary differential equations (ODEs) while simultaneously extracting multiscale features. Then, CD is performed under superpixel enhancement to improve the homogeneity of the CD maps. Diffpurifier not only integrates IT and feature extraction, simplifying the workflow, but also maintains high accuracy, stable training, and generalization to different types of data without the need for additional translation constraints. In comparative experiments on four public datasets, Diffpurifier outperforms the second-best method by an average of approximately 5% in terms of $F1$-score, validating the effectiveness and robustness of the method.**

***Index Terms*— Change detection (CD), denoising diffusion probabilistic models (DDPMs), image translation (IT), optical, ordinary differential equations (ODEs), synthetic aperture radar (SAR).**

I. Introduction

NATURAL and man-made disasters continuously pose threats to human life and property safety, and unfortunately, with the intensification of global climate change, the frequency and intensity of extreme weather events and natural disasters are increasing. Although humans currently do not have the ability to completely prevent disasters from occurring, we can greatly reduce the damage caused by disasters to humans through early warning systems, monitoring during disasters, and post-disaster rescue and reconstruction efforts. Therefore, research on disaster monitoring and early warning systems is of paramount importance.

Over the past two decades, the rapid growth of remote sensing technology has provided an increasing number of geo-spatial information products through continuous monitoring and data collection over large areas [1]. Particularly in the field of disaster management, remote sensing technology has become one of the most important tools. Using remote sensing images for change detection (CD) can not only obtain real-time information about the situation in disaster areas but also provide important loss assessment data after disasters, providing a scientific basis for emergency response and recovery work.

Optical images are commonly used for CD due to their periodic acquisition and consistency with human visual characteristics. However, disasters are often accompanied by extreme weather conditions, making it difficult to obtain clear, cloud-free optical images. With the development of various sensor platforms [2], synthetic aperture radar (SAR) has become the preferred tool for acquiring disaster damage information through remote sensing, owing to its all-weather, day-and-night imaging capabilities. As a result, our research focuses on the CD between pre-disaster optical images and post-disaster SAR images, aiming to improve the efficiency and accuracy of disaster response.

One major challenge in this area is the significant image differences caused by the distinct imaging mechanisms of the two sensors, which makes direct comparison difficult. Currently, the basic idea of heterogeneous CD is to convert multisource data into a common space. Based on the type of transformed space, mainstream methods can be categorized into three types.

The category transformation (CT)-based method classifies heterogeneous images based on the categories of land features and then compares the categorized images to achieve detection results [3]. The advantage is that it can minimize differences such as atmospheric conditions, lighting, and sensor variations, and the results can directly display the categories of the changing areas, reflecting complete information about the changes in land features. However, CD results are susceptible to cumulative categorization errors, which require high precision and expertise in the preliminary operations. Nowadays, such methods are moving toward combining classification and detection, where compound classification is a feasible option but still in the exploratory stage [4], [5].

The feature transformation (FT)-based method translates multitemporal images into a common feature space using manual or automatic feature extraction, allowing their features

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to be directly compared [6]. Traditional methods use machine learning (ML) or statistical methods such as HPT [7], PP [8], AM-IR [9], while deep learning (DL)-based methods, like SCCN [10], MC-CSSL [11], DAHCD [12], X-Net [13], use deep belief networks (DBNs), convolutional neural networks (CNNs), etc., to extract high-level features. These highly flexible methods can handle high-dimensional images but ignore the overall spatial relationships of the images [14], and may suffer from loss of original image information and degradation of deep features during the optimization [6].

The image translation (IT)-based approach utilizes a homology method to detect changes after translating heterogeneous images from one domain to another. Once IT is completed, various well-established homology CD methods can be applied to obtain the final results. This approach aligns more with human visual perception and avoids the complexity of feature space alignment. Additionally, it preserves more useful original modality information while reducing CD errors caused by modality differences, such as those from imaging angles.

With the advancement of generative models and IT methods, IT-based methods have gradually become the main trend. However, there still exist some concerns.

1) *Redundant feature extraction:* Many existing methods adopt a two-stage strategy in their overall design, where features are extracted after the IT step is completed. Image features can easily degrade during propagation, leading to the loss of original information and a reduction in detection accuracy.
2) *Poor generalization ability:* These algorithms typically use generative adversarial networks (GANs) [15] to perform IT, which requires adversarial training. The adversarial training between the generator and discriminator makes convergence difficult and often results in mode collapse [16], [17]. On the other hand, non-GAN methods often involve complex loss functions, such as weighted translation loss, cycle-consistency loss, reconstruction loss, and weight decay regularization terms. In practical applications, both methods require manual intervention to adjust hyperparameters when transferred to new datasets to balance the effectiveness of IT and the accuracy of CD.
3) *Incomplete and broken results:* The objects in the CD map of these methods are usually incomplete. This may be because the IT model fails to accurately identify noise in the input image, thus transmitting the noise to the detection results. Alternatively, if the translation model fails to correctly learn the features of the target domain or the training is unstable, it may also result in broken detection objects.

Recently, the demand for AI-generated content (AIGC) has driven the research on diffusion models. Compared with GANs that require adversarial training, denoising diffusion probabilistic models (DDPMs) perform better in generating diversity and training stability, and are therefore increasingly being used for many tasks such as conditional generation [18], [19], [20] and IT [21], [22]. In the field of remote sensing, many works utilize the powerful ability of diffusion models to learn image distribution and extract key semantics to perform many downstream tasks. For example, DDPM-CD [23] retrieves the multiscale features of the decoder and a training classifier head for homologous CD. DiffUCD [24] employs a diffusion model to retrieve semantically relevant features from hyperspectral imagery. CADM [25] utilizes the differences between multilevel features to guide the sampling process in the detection of optical images.

Witnessing the excellent performance of the diffusion model in homologous CD tasks, we tried to apply it to optical and SAR image CD. Therefore, considering the high demand for model generalization and high precision in CD for disaster emergency scenarios, we propose an innovative DDPMs-based model for optical and SAR CD, which integrates IT and feature extraction. By leveraging the advantages of diffusion models [26], which can extract key semantic features from input images and generate multilevel feature representations [23], we extract multiscale features from multisource images during IT. This not only retains and transfers the key semantic information of the original image but also enhances the model's expression and adaptability to different types of data and image domains. Furthermore, by using superpixel-pixel associations, we annotate the change results on the detection map with the region block information from both time periods, improving the homogeneity of the detection results. Compared to previous methods, our algorithm does not need additional translation constraints, reducing the dependency of detection results on specific tasks and simplifying the overall workflow. Experimental results on publicly available datasets demonstrate that our method achieves high precision while keeping strong generalization capabilities.

The main contributions of our work are as follows.

1) A novel Diffpurifier is proposed to integrate IT and feature extraction for optical and SAR CD tasks. This method not only simplifies the overall workflow and prevents degradation of image features during propagation but also ensures stable training, high accuracy, and strong generalization ability.
2) Superpixel-pixel associations are employed to enhance the homogeneity of detection results. Through this combination, the block information of the bitemporal images is added to the change results, enhancing the homogeneity of the detection results and improving the accuracy of the algorithm in detecting change areas.
3) The experiments on four publicly available optical and SAR CD datasets demonstrated the excellent performance of our model, with an average $F1$-score of about 5% higher than the second-best method for detecting change areas. Moreover, it is demonstrated that a better CD model can still produce good results without relying on translation task constraints, which is valuable for transferring optical and SAR CD tasks to more new sensors in the future.

The rest of the article is organized as follows. In Section II, we summarized some state-of-the-art IT-based methods and introduced some basic knowledge related to diffusion models. In Section III, the details of our network are presented, followed by experiments and results in Section IV. The

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discussion is presented in Section V. Finally, we draw the conclusion in Section VI.

II. Preliminaries

A. IT-Based Methods for Heterogenous CD

IT is the process of assigning attributes to another domain while preserving the content of an image, with the aim of learning the mapping relationship from the source image space to the target image space [27]. With the advancement of technology, many GAN-based methods have been developed to implement this concept. Typically, GANs consist of a generator and a discriminator. The generator aims to create realistic target domain images, while the discriminator's role is to distinguish between generated images and real images. During the training process, the generator and discriminator engage in a game-like competition, with the generator ultimately learning to produce samples that closely resemble the real target domain image distribution, thereby enabling the transformation of source domain images into target domain images.

In remote sensing image CD tasks, many methods utilize GANs and their variants to perform IT: based on conditional GANs (cGANs) [28], Niu et al. [29] fed optical images to the generator with additive Gaussian noise and designed an approximation network to reduce pixel-wise differences between the original SAR image and the translated one. Subsequently, the cycle-consistency adversarial networks (cycleGANs) [30] proposed the latent space hypothesis, which suggests that corresponding images from different domains can be mapped to the same latent variables in a shared latent space, addressing the issue of a lack of paired examples. This inspired Liu et al. [14] to complete heterogeneous CD tasks in an unsupervised manner. In addition, Chen et al. [31] pretrained cycleGAN to obtain two generators that can obtain prior information on transforming between two distributions. On the basis of cycleGANs, the no-independent-component-for-encoding GAN (Nice-GAN) [32] was more compact and efficient and adopted in FRAN [33], DTCDN [34], and MTCDN [35]. Especially, the last algorithm incorporates the CD operation into the GANs based on DTCDN. What's more, Radoi [36] built GANs over the U-Net architecture to achieve intermodality translation, which can yield more accurate translated image and also got impressive CD result.

Despite the improvements in GANs and GAN-based CD methods, we must acknowledge that training GANs is still a rather challenging task. In practical applications, issues such as training instability, vanishing gradients, and mode collapse are common when dealing with new datasets, leading to significant waste of time and computational resources. Moreover, due to the training strategy, IT and feature extraction for CD operations must be treated as separate tasks, resulting in redundant steps.

Although there are also non-GAN-based methods, such as ACE-Net [13] using autoencoders (AE) and X-Net [13] using fully convolutional networks, these methods are not exempt from the complexity of loss functions, such as translation loss, cycle-consistency loss, and reconstruction loss. This makes them less effective when applied to new datasets.

Therefore, under the premise of stable training, ensuring accuracy and improving generalization capability are key directions for enhancing models in emergency scenarios.

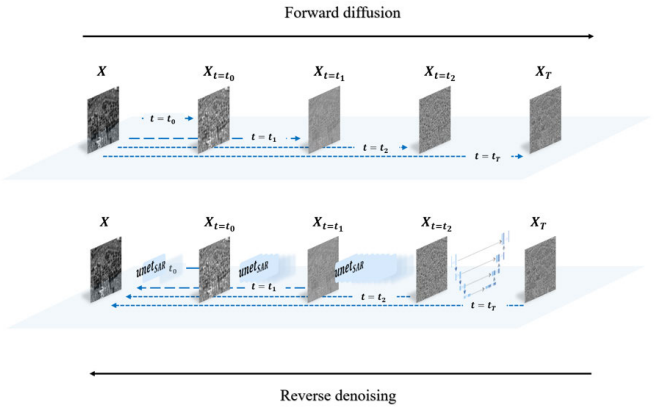


Fig. 1. Diffusion training process.

B. Diffusion Model and Differential Equations

As illustrated in Fig. 1, the diffusion model is trained based on reversibility to simulate the process of reconstructing the original clean image from noise, consisting of two stages: forward diffusion and reverse denoising [26].

In the forward process, following a Markov chain with a total timestep of T, Gaussian noise $\epsilon = \mathcal{N}(0, I)$ is gradually added to the clean image x_0 until the image is completely destroyed. The image distribution at timestep t can be defined as follows:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}\right) \tag{1}$$

where $t = 1, 2, \ldots, T$, β_t is the Gaussian variance from a noise schedule $\beta_1, \beta_2, \ldots, \beta_T$ and can be expressed as follows because of the additivity of Gaussian noise:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, \left(1-\bar{\alpha}_t\right)\mathbf{I}\right) \tag{2}$$

where $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. Unfold the above distribution can obtain the image at timestep t

$$\mathbf{x_t} = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon. \tag{3}$$

The reverse process is a parameterized Gaussian distribution

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)) \tag{4}$$

and according to variational inference, the ultimate optimization goal of the diffusion model is to minimize the KL divergence of the two distributions $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$. By using a fixed variance and the reparameterization trick, the optimization objective of the diffusion model can be simplified to predict the noise from x_{t-1} to x_t

$$\mathbb{E}_{x_0,\epsilon}\left\|\epsilon_\theta^{(t)}(x_t) - \epsilon\right\|_2^2. \tag{5}$$

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Thus, once the noise predictor $\epsilon_\theta^{(t)}(x_t)$ is trained, the sample x_0 can be iteratively reconstructed from latent variables.

Furthermore, from a mathematical perspective, both the forward and backward diffusion processes can be represented by stochastic differential equations (SDEs). Specifically, when making the DDPM forward process continuous by expressing the discrete β_t as $\beta(t)\Delta t$, the forward process can be written as

$$\mathbf{x_t} = \sqrt{1-\beta(t)\triangle t}\mathbf{x_{t-1}} + \sqrt{\beta(t)\triangle t}\mathcal{N}(0,\mathbf{I}). \tag{6}$$

For sufficiently small timestep, performing a first-order Taylor expansion yields the approximate solution

$$\mathrm{x_t} \approx \mathrm{x_{t-1}} - \frac{\beta(\mathrm{t})\triangle \mathrm{t}}{2}\mathrm{x_{t-1}} + \sqrt{\beta(\mathrm{t})\triangle \mathrm{t}}\mathcal{N}(0,\mathrm{I}). \tag{7}$$

This is essentially the Euler–Maruyama discretization of an Itô diffusion SDE

$$\mathrm{d}x_t = -\frac{1}{2}\beta(t)x_t\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\omega_t. \tag{8}$$

$\mathrm{d}\omega_t$ corresponds to the Brownian motion increment in the reverse process. The reverse process of this SDE is also an SDE, and its specific form is

$$\mathrm{d}x = -\frac{1}{2}\beta(t)[\mathbf{x_t} + 2\nabla_{\mathbf{x_t}}\log p_{\mathbf{t}}(\mathbf{x})]\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\omega_t. \tag{9}$$

Song et al. [37] further proved that any SDE can be converted into a probability flow ODE (PF ODE) preserving the same marginal distribution

$$\mathrm{d}\mathbf{x_t} = \left[f(\mathbf{x_t},t) - \frac{1}{2}g(t)^2\nabla_{\mathbf{x_t}}\log p_t(\mathbf{x_t})\right]\mathrm{d}t. \tag{10}$$

Here, $f(\mathbf{x_t},t)$ is the drift term representing the deterministic trend, $g(t)$ is the diffusion term representing the strength of random perturbations, and $\log p_t(\mathbf{x_t})$ is the score function, i.e., the gradient of the log probability density.

Expressing the diffusion and sampling processes in the form of differential equations allows us to view the latent space of the diffusion model composed of $X_1, X_2, \ldots, X_T$ as a Gaussian distribution space with a continuous trajectory structure. By leveraging SDEs or deterministic ordinary differential equations (ODEs), images from different modalities can be mapped to distinct locations in the latent space. Through numerical integration, the forward diffusion and reverse generation processes are reformulated as tractable computations.

In contrast to the stochastic generation process in SDEs, the ODE formulation removes random noise terms $\sqrt{\beta(t)}\mathrm{d}\omega_t$, resulting in a deterministic and reversible generation path. The forward and reverse processes are described by the same ODE but with opposite integration directions: forward integration maps observed images to their latent representations, whereas reverse integration corresponds to the image generation process.

III. Methodology

In this section, we present the proposed optical and SAR CD method based on diffusion purification. As shown in Fig. 2, our method can be divided into two parts: diffusion-based feature purification and superpixel-enhanced CD. First, by leveraging pre-trained diffusion models in combination with ODEs, we simultaneously perform IT and extract multiscale features from multisource images. This joint design not only simplifies the overall workflow but also enhances the model's adaptability and transferability across different tasks. Next, by using superpixel-pixel associations, we incorporate information from bitemporal region blocks to refine the change map, thereby improving the spatial consistency and homogeneity of the detection results. Our algorithm does not impose additional constraints on the translation process and employs a simple loss function, which makes the model training more stable and less sensitive to hyperparameters, enabling it to adapt to various new datasets.

A. Diffusion-Based Feature Purification

Diffusion models have demonstrated remarkable capabilities in learning image distributions. A critical aspect of adapting them to downstream tasks lies in controlling the generative process to produce specific, deterministic samples. We achieve this by manipulating the latent space of the model, a high-dimensional space composed of $X_1, X_2, \ldots, X_T$, which governs the generation dynamics.

Specifically, we independently train a diffusion model for each image domain and apply them to perform translation and reconstruction of coregistered pre-event optical and post-event SAR images. By integrating the decoder outputs across multiple noise levels, we construct a comprehensive representation composed of multiscale features extracted at different timesteps.

1) Pre-Training of Diffusion Models: In Diffpurifier, we first train two independent unconditional diffusion models (DDPMs) in the optical and SAR domains, respectively, fixing their parameters upon completion of training.

During the sampling phase, we employ the DDIM sampler, which maintains the same training mechanism as DDPM while constructing a non-Markovian diffusion process to enable skip-step sampling [38]. DDIM significantly reduces the number of inference steps while preserving generation quality, with its iterative process represented as follows:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta^{(t)}(x_t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_{t-1}}\cdot\epsilon_\theta^{(t)}(x_t). \tag{11}$$

More importantly, DDIM belongs to the category of deterministic generation: Given the same input and parameters, the model produces identical results each time. From a mathematical perspective, its iteration can be rewritten as

$$\frac{x_{t-1}}{\sqrt{\bar{\alpha}_{t-1}}} = \frac{x_t}{\sqrt{\bar{\alpha}_t}} + \left(\sqrt{\frac{1-\bar{\alpha}_{t-1}}{\bar{\alpha}_{t-1}}} - \sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t}}\right)\epsilon_\theta^{(t)}(x_t). \tag{12}$$

By reparameterization

$$\sigma(t) = \sqrt{\frac{1-\bar{\alpha}(t)}{\bar{\alpha}(t)}}, \quad \bar{\mathbf{x}}(t) = \frac{\mathbf{x}(t)}{\sqrt{\bar{\alpha}(t)}}. \tag{13}$$

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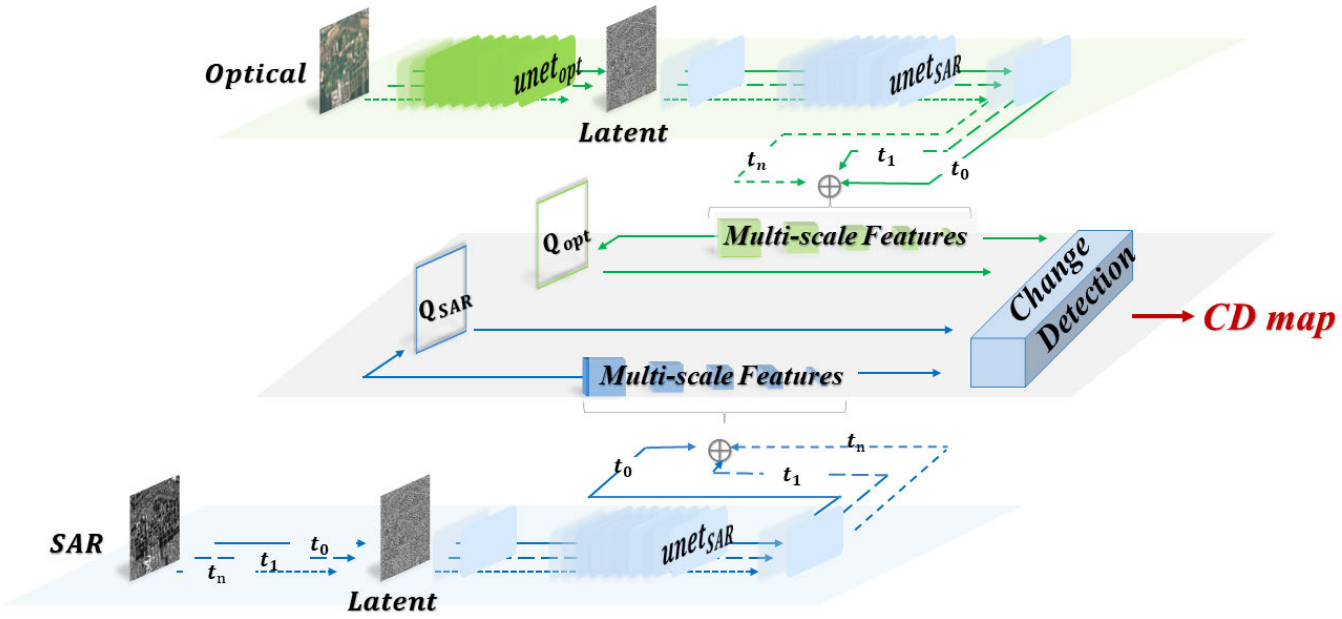


Fig. 2. Overall framework of the proposed network. Please note that the mapping between image domains by the diffusion model requires the use of ODEs, which are omitted in the figure.

Equation (12) can be regarded as the solution of the following ODE:

$$\mathrm{d}\bar{\mathbf{x}}(t) = \epsilon_\theta^{(t)}\left(\frac{\bar{\mathbf{x}}(t)}{\sqrt{\sigma^2+1}}\right)\mathrm{d}\sigma(t). \tag{14}$$

The iterative process of DDIM, i.e., (11), can be regarded as the backward solution of (14). While the forward solution of (14), i.e., the integration process from $t = 0$ to $t = T$, represents the mapping from the clean image to the latent space.

2) ODE-Based IT: The core challenge in IT lies in maintaining content consistency while separating style differences. In our method, the key to unconstrained translation lies in using the probability flow ODE to establish an invertible mapping between image space and latent space.

Specifically, we adopt the dual diffusion implicit bridges (DDIBs) algorithm [39] to decompose the IT process into two sequential stages: from source domain images to latent representations, then from latent representations to target domain images.

First, the optical domain diffusion model ϵ_{opt} encodes the optical image X_0 into latent space by solving forward ODE

$$X_l = X_0 + \int_{t=0}^{T} \epsilon_{\text{opt}}(X_t; t)\mathrm{d}t. \tag{15}$$

Then, the SAR domain diffusion model ϵ_{SAR} decodes the latent representation X_l into the target domain by solving backward ODE (i.e., DDIM sampler)

$$X_{\text{SAR}} = X_l + \int_{t=T}^{0} \epsilon_{\text{opt}}(X_t; t)\mathrm{d}t. \tag{16}$$

Notably, although the forward ODE integration aligns in direction with the forward diffusion process (from $t = 0$ to $t = T$), they are fundamentally different: the former is a deterministic trajectory preserving semantic information, while the latter is a stochastic perturbation used only during training.

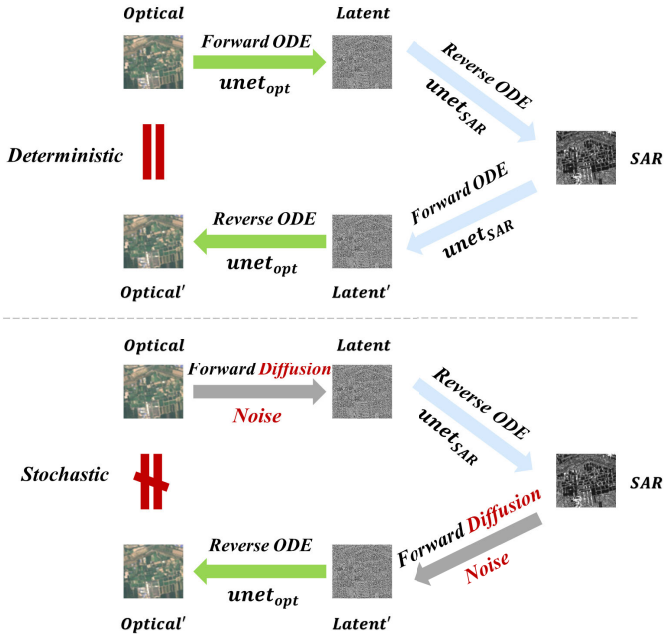


Fig. 3. Comparison of latent variable acquisition methods. The ODE-based deterministic path inherently satisfies cycle consistency in IT, requiring no extra alignment constraints. In contrast, forward diffusion introduces stochasticity and thus necessitates additional translation supervision.

Fig. 3 provides a comparison between different approaches to obtaining latent representations. As demonstrated by Su et al. [39], the deterministic path defined by ODEs inherently satisfies the principle of cycle consistency in IT tasks, thus

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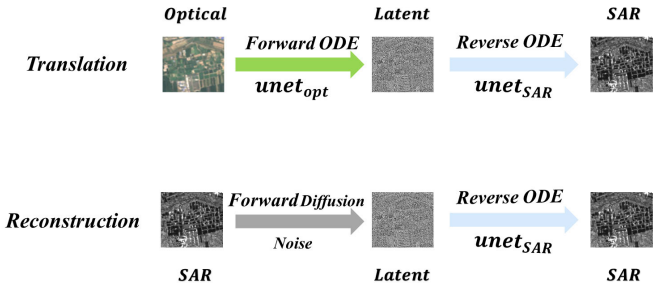


Fig. 4. Asymmetric design of diffusion paths for optical and SAR images.

eliminating the need for additional translation constraints or explicit alignment of the latent space during training. In contrast, directly mapping images to the Gaussian space via the forward diffusion process introduces stochastic noise, which inevitably leads to information loss and necessitates the use of a source domain reconstruction network to compensate for it.

From an information-theoretic perspective, the ODE-based mapping transforms the relationship between the input image and the latent variable into a bidirectional and lossless transformation. As a result, the latent representation inherently preserves sufficient semantic and structural information to reconstruct the original image. Consequently, we interpret the high-dimensional latent space of the diffusion model as a shared semantic space that stores high-level, modality-invariant abstractions. This space serves as a semantic bridge, while the diffusion model itself is responsible for decoupling semantics from modality-specific styles.

3) Integration of Multiscale and Multitimestep Features: To further unify IT and feature extraction, we exploit the multiscale representation capabilities of the diffusion model decoder.

As DDPM-CD [23] verifies, the diffusion model decoder implicitly captures hierarchical structural and semantic representations at different stages. Hence, during the translation path from optical to latent to SAR, we extract intermediate features from the SAR domain decoder to obtain hierarchical representations of multiple receptive fields.

For post-event SAR images Y_0 that are already in the target domain, we apply a self-reconstruction approach to extract symmetrical structural features. Since no cross-domain mapping is involved, cycle consistency is not required. Thus, we directly apply forward diffusion [i.e., (3)] to obtain the latent code

$$Y_l = \sqrt{\bar{\alpha}_t} Y_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{17}$$

and reconstruct it using the SAR domain diffusion model

$$Y_{\text{SAR}} = Y_l + \int_{t=T}^{0} \epsilon_{\text{SAR}}(Y_t; t)\mathrm{d}t. \tag{18}$$

This process is a natural extension of the diffusion training mechanism and avoids unnecessary cross-domain error propagation while ensuring that the final features for CD share consistent semantic granularity and structural expressiveness.

Fig. 4 shows the asymmetric design of diffusion paths for optical and SAR images, which not only accounts for the semantic differences between domains but also fully exploits the representational power of diffusion decoders.

Considering that in the diffusion model training process, the timestep t corresponds to the progressive intensity of the noise perturbation over time, smaller values of t represent weaker noise, where the image retains more fine-grained details and the model exhibits stronger noise discrimination. In contrast, larger t values indicate stronger noise perturbations; although more abstract semantic information may emerge, the model's robustness to noise tends to degrade.

Based on this observation, we control both the magnitude and the number of selected timesteps t to flexibly adjust the depth and frequency of feature purification. This enables us to extract feature representations under different purification intensities, denoted as ${X_{\text{SAR}}}^{i,t}$, ${Y_{\text{SAR}}}^{i,t}$, where $i = 0, 1, 2, 3, 4$ and $t = t_1, t_2, \ldots, t_n$.

Subsequently, a pixel-wise addition is employed to fuse multiscale features across different timesteps, enhancing the stability and reliability of the final representation

$$X^i_{\text{SAR}} = X^{i,t_1}_{\text{SAR}} + X^{i,t_2}_{\text{SAR}} + \cdots + X^{i,t_n}_{\text{SAR}} \tag{19}$$

$$Y^i_{\text{SAR}} = Y^{i,t_1}_{\text{SAR}} + Y^{i,t_2}_{\text{SAR}} + \cdots + Y^{i,t_n}_{\text{SAR}}. \tag{20}$$

This mechanism enriches feature diversity while preserving semantic integrity, thereby enhancing the model's generalization capability. As illustrated in Fig. 5, we use the optical image branch as an example, where different colored lines denote features extracted from different timesteps. This fusion strategy effectively integrates multilevel structural and semantic information, providing a robust and informative representation for downstream CD tasks. Algorithm 1 shows the process of this section.

Algorithm 1 Procedure of Diffusion-Based Feature Purification

Input:
X_0: optical image;
Y_0: SAR image;
$t_1, t_2, \ldots, t_n$: timesteps;
ϵ_{opt}: pre-trained optical domain diffusion model;
ϵ_{SAR}: pre-trained SAR domain diffusion model

Output:
$X_{SAR}^{i=0,1,2,3,4}$: multi-scale translated optical features;
$Y_{SAR}^{i=0,1,2,3,4}$: multi-scale SAR features;

for $t = t_1$ *to* t_n **do**
  `// Optical IT`
  Obtain latent variables of optical images with (15);
  Mapping latent variables to SAR image domain with (16);
  `// SAR image reconstruction`
  Obtain latent variables of SAR images with (17);
  Reconstruct clean SAR images with (18);
end
Integrate multi-timestep features with (19) and (20);

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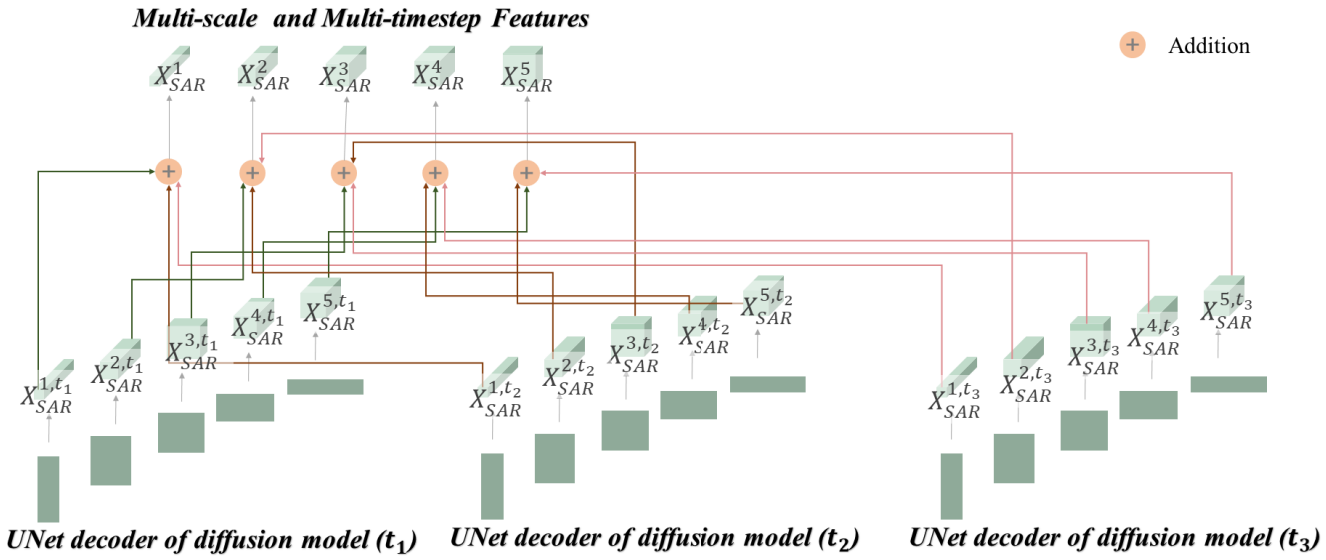


Fig. 5. Multiscale and multitimestep feature integration. Lines in different colors indicate features extracted at different timesteps.

B. Superpixel-Enhanced CD

Fig. 6 illustrates the overall architecture of the CD network. The left pixel-level branch serves as the primary pathway for extracting change information from multilevel bitemporal features. The right superpixel-level branch enhances the homogeneity of the detection results and promotes boundary alignment with actual ground objects. Finally, the ensemble channel attention module (ECAM) is applied to perform information selection and fusion between the superpixel-level features (SP) and pixel-level features (PF), automatically emphasizing the most relevant and effective change-related information from both sources.

1) Superpixel-Pixel Association Matrices Acquisition: The superpixel–pixel association matrix Q characterizes the relationship between pixels and superpixels in the image, where each row corresponds to a specific superpixel. With Q, bidirectional mapping can be performed between pixel-level features I and superpixel-level features S. We adopt the differentiable simple linear iterative clustering (SLIC) algorithm [40] as the core clustering method, which initializes with a regular grid and iteratively updates the superpixel features. The number of superpixels M and the number of iterations J are fixed.

Specifically, before the iterations begin, the feature maps I (i.e., X^4_{SAR} and Y^4_{SAR}), having the same spatial resolution as the original images, are divided into $M_h \times M_w$ regular grids. Within each grid, the mean of pixel features is computed to initialize the superpixel features

$$S^0_{M\times K} = \mathrm{Avg}(I). \tag{21}$$

In the jth iteration, the soft association between the pth pixel and the qth superpixel is calculated as

$$Q^{\mathrm{j}}_{p,q} = e^{-|I_p - S^{j-1}_q|^2}. \tag{22}$$

Then, the superpixel cluster centers are updated as

$$Z^j_i = \sum_p Q^j_{pi} \tag{23}$$

$$S^j_i = \frac{1}{Z^j_i}\sum_{p=1}^{n} Q^j_{pi} I_p. \tag{24}$$

The association matrix Q describes the correspondence between pixels and superpixels. By column-normalizing Q (denoted as $\hat{Q}$), pixel-level features I can be mapped to the superpixel space

$$S = \hat{Q}^T I. \tag{25}$$

Alternatively, row-normalization of Q (denoted as $\tilde{Q}$) allows reconstruction of pixel-level features from superpixel features

$$I = \tilde{Q} S. \tag{26}$$

This iterative process resembles the EM algorithm, alternating between updating Q and S via the above equations. Ultimately, we obtain the superpixel–pixel association matrices Q_{opt} and Q_{SAR} for the optical and SAR images, respectively.

2) Pixel-Level Change Feature Acquisition: The pixel-level branch serves as the main structure of the CD network, modified from SNUNet [41]. It adopts a densely connected residual decoder architecture to fuse multilevel features for change information extraction, while upsampling feature maps to the original image dimensions. The architecture is illustrated in Fig. 6, where numerical annotations indicate the output channels of each feature map.

Initially, multiscale dual-temporal features generated by the diffusion model at different timesteps are reduced in dimensionality using 1×1 convolutions, resulting in feature maps with 32, 64, 128, 256, and 512 channels, respectively, to reduce computational complexity.

Subsequently, upsampling is conducted via a decoder comprising four-stage decoding units, with each unit taking inputs

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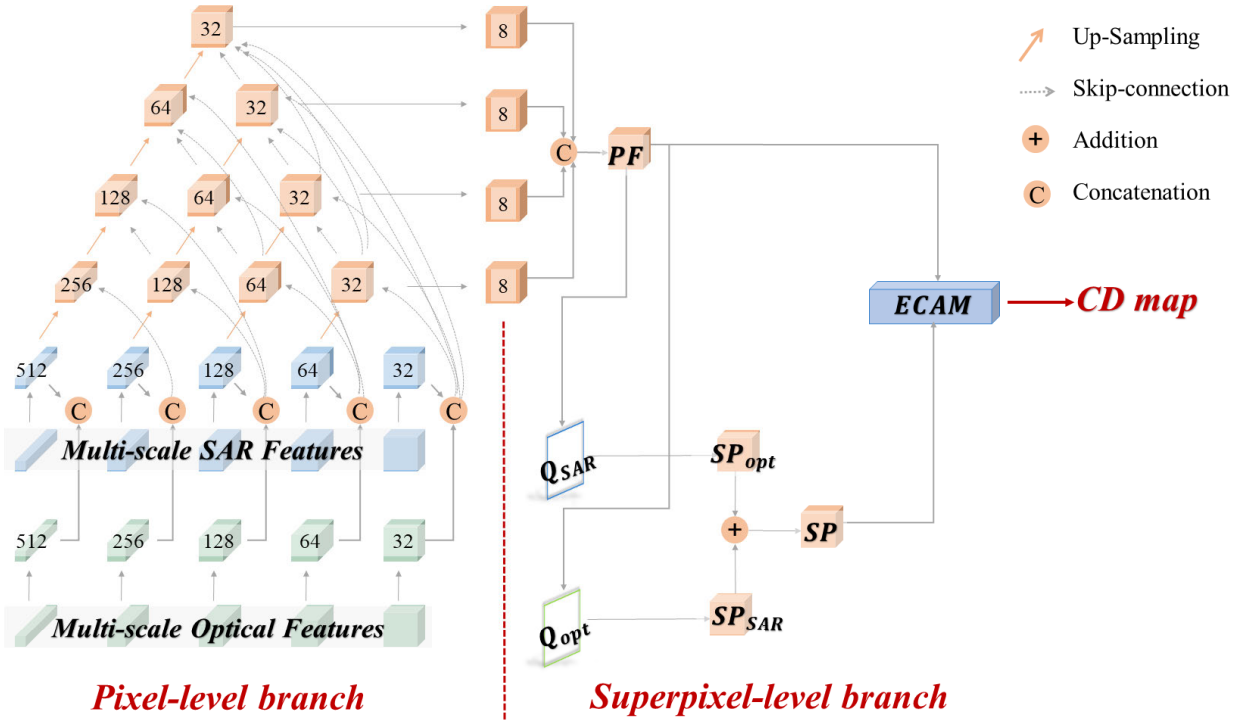


Fig. 6. Overall architecture of the CD network.

from both the preceding-stage decoding unit and its same-stage counterpart.

Each decoder unit consists of a residual double convolution block and an upsampling module. The input is formed by concatenating the dual-temporal features and the upsampled features from the previous decoder. These are then fed into a 3×3 double convolution block for feature extraction. As shown in Fig. 7, each convolution layer is followed by a pooling layer and ReLU activation, and residual connections are employed to facilitate gradient flow and accelerate convergence.

Skip connections between decoders help integrate shallow spatial details with deep semantic features and alleviate the loss of spatial localization in deeper layers. The final decoder outputs four feature maps at the original resolution, capturing changes at multiple semantic levels.

These four maps are concatenated along the channel dimension and processed through a 1×1 convolution to obtain the final eight-channel pixel-level feature map, denoted as PF

$$\mathrm{PF} = \mathrm{Decoders}\left(X_{\mathrm{SAR}}^{i=0,1,2,3,4}, Y_{\mathrm{SAR}}^{i=0,1,2,3,4}\right). \tag{27}$$

3) Superpixel-Level Change Feature Acquisition: Traditional pixel-level methods are prone to noisy detections and blurry boundaries when handling SAR speckle noise and high-resolution imagery. To address this, a superpixel-level branch is introduced to enhance spatial consistency and boundary alignment via over-segmentation. This branch consists of three steps.

First, use the superpixel–pixel association matrices Q_{opt} and Q_{SAR} to project the pixel-level feature map PF into the superpixel space

$$F_{\mathrm{opt}} = \hat{Q}_{\mathrm{opt}}^{T}\mathrm{PF}, \quad F_{\mathrm{SAR}} = \hat{Q}_{\mathrm{SAR}}^{T}\mathrm{PF}. \tag{28}$$

Here, $Q \in \mathbb{R}^{M\times N}$, $\mathrm{PF} \in \mathbb{R}^{C\times N}$, $F \in \mathbb{R}^{\mathrm{CD}}$. This projection aggregates the features of pixels within each superpixel, reducing the impact of noisy pixels in PF. Since Q_{opt} and Q_{SAR} are derived from the deep features of bitemporal images, they encode different boundary-aware representations, enhancing edge sensitivity during fusion. This step functions as a superpixel-guided pooling operation to denoise and smooth intra-object variations.

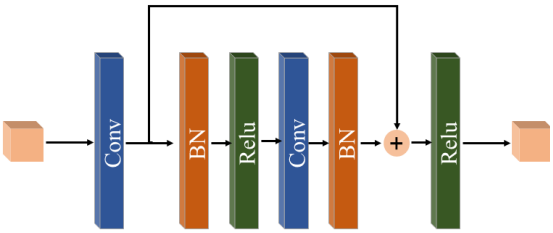


Fig. 7. Double convolution block.

The second step involves remapping the superpixel feature representations back to the pixel space, which is again implemented using the superpixel-to-pixel association matrices Q_{opt} and Q_{SAR}

$$\mathrm{SP}_{\mathrm{opt}} = \widetilde{Q}_{\mathrm{opt}} F_{\mathrm{opt}}, \quad \mathrm{SP}_{\mathrm{SAR}} = \widetilde{Q}_{\mathrm{SAR}} F_{\mathrm{SAR}}. \tag{29}$$

The goal is to realign the superpixel features of the bitemporal phases for pixel-wise fusion. From a model design perspective, this step is symmetric to the previous one: the former pools pixel features to superpixels, while this step unpools them

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back. Equations (28) and (29) can be unified into an end-to-end operation

$$\mathrm{SP}_{\mathrm{opt}} = \widetilde{Q}_{\mathrm{opt}} \hat{Q}_{\mathrm{opt}}^{T} \mathrm{PF}, \quad \mathrm{SP}_{\mathrm{SAR}} = \widetilde{Q}_{\mathrm{SAR}} \hat{Q}_{\mathrm{SAR}}^{T} \mathrm{PF}. \tag{30}$$

This process forms an aggregation-to-expansion encoding–decoding structure, similar to attention mechanisms in Transformers or AEs. When Q is not an identity matrix and $M < N$, information loss occurs during the mapping and unmapping, which contributes to the denoising and smoothing effects of the superpixel branch.

The final step involves fusing the remapped pixel-wise features $\mathrm{SP}_{\mathrm{opt}}$ and $\mathrm{SP}_{\mathrm{SAR}}$ to obtain the superpixel-level change feature that integrates boundary information from both time phases. In this work, we follow the procedure of ESCNet [42] and combine $\mathrm{SP}_{\mathrm{opt}}$ and $\mathrm{SP}_{\mathrm{SAR}}$ through element-wise addition. A weighted strategy can also be applied to balance the importance of the bitemporal images

$$\mathrm{SP} = \mathrm{SP}_{\mathrm{opt}} + \mathrm{SP}_{\mathrm{SAR}}. \tag{31}$$

4) ECAM Between Pixel-Level and Superpixel-Level Change Features: The superpixelization operation can enhance the consistency of features within a local region, thereby suppressing the interference of potential noise on the detection results and reducing the generation of false changes. However, directly determining the change regions based on the output of the superpixel module may lead to two problems.

On the one hand, current object segmentation methods generally struggle to ensure that the segmented regions strictly correspond to real object boundaries, which is particularly evident under superpixel-based frameworks. This is because superpixels represent an over-segmentation of the image, which is designed to preserve object edge structures as much as possible. However, within the same object, multiple superpixels may exhibit irregular shapes and loose boundaries.

On the other hand, in remote sensing CD tasks, changes are often caused by the appearance or disappearance of certain objects at different time points. Therefore, accurately delineating the object boundary in the temporal phase where the object is absent is extremely challenging.

The above factors may lead to misalignment between superpixel regions from the bitemporal phases, ultimately resulting in blurred boundary information in the fused results. Although the superpixel mechanism shows certain advantages in enhancing the features of each temporal phase, its underlying principle does not possess the capability to align structural information across different phases. Thus, it needs to be combined with pixel-level feature processing to recover the object boundary information that is weakened during temporal fusion.

Therefore, we apply an ECAM [41] between the superpixel-level change feature SP and the pixel-level change feature PF to automatically select and focus on the most relevant information between these two levels of change features. Specifically, ECAM can be expressed as

$$\begin{aligned} \mathrm{CAM} = {} & \sigma(\mathrm{MLP}(\mathrm{MaxPool}(X)) \\ & + \sigma(\mathrm{MLP}(\mathrm{AvgPool}(X)) \end{aligned} \tag{32}$$

$$\mathrm{MF} = [\mathrm{SP}, \mathrm{PF}] \tag{33}$$

$$\mathrm{ECAM} = \left(\mathrm{MF} + \mathrm{repeat}_{(2)}(\mathrm{CAM}(\mathrm{SP} + \mathrm{PF}))\right) \otimes \mathrm{MF}. \tag{34}$$

The purpose of this step is to reconstruct the boundary regions blurred during the fusion process, and, on the other hand, to identify subtle change areas that are missed due to superpixel segmentation.

Finally, the network outputs a two-channel change probability map, which represents the change conditions of the target regions. The process of this section is shown in Algorithm 2. During training, we only use focal loss [43] as the loss function to alleviate the impact of class imbalance between the ground-truth image and the ECAM output. This eliminates the need for translation constraints in our model, making it no longer reliant on the detection results from the translation task.

Algorithm 2 Procedure of Superpixel-Enhanced Multi-scale CD

Input:
$X_{SAR}^{i=0,1,2,3,4}$: multi-scale translated optical features;
$Y_{SAR}^{i=0,1,2,3,4}$: multi-scale SAR features;
N: number of pixels;
M: number of superpixels;
K: channel of superpixels;
J: Number of iterations

Output:
CM: CD map

Initialize superpixel centers with (21);
for *each iteration* $j = 1$ *to* J **do**
  Compute association between each pixel p and the surrounding superpixel i with (22);
  Compute new superpixel centers with (23) and (24);
end
Pixel-level change feature generation with (27);
Superpixel-Guided Pooling with Eq. (28);
bitemporal superpixel change feature acquisition with Eq. (29);
bitemporal superpixel change feature fusion with Eq. (31);
ECAM between pixel-level and superpixel-level change features with Eq. (32), Eq. (33) and Eq. (34);

IV. Experiments

First, we introduce the four publicly available datasets and experimental settings used in our work. Then, the comparison methods and evaluation metrics are briefly summarized. Finally, we present the qualitative and quantitative evaluation results for each dataset.

A. Dataset Description and Hyperparameter Setting

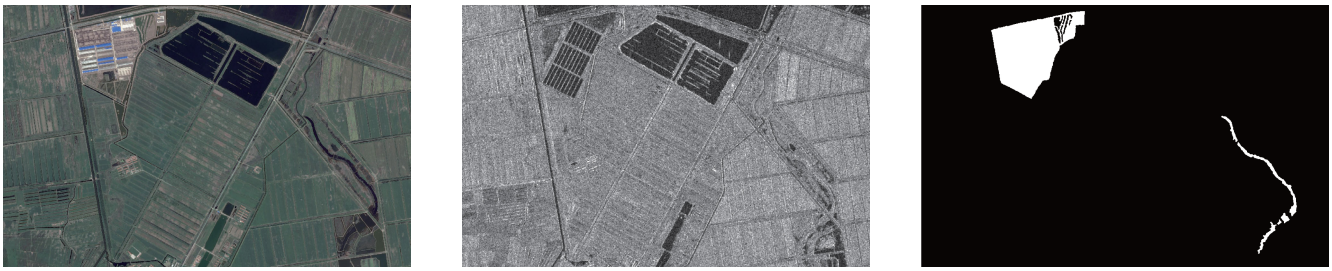
We conducted experiments on four available benchmark datasets. Each of them contains a pair of coregistered optical and SAR images and the corresponding ground truth. The detailed information is shown in Table I.

1) *Shuguang dataset:* As shown in Fig. 8, the first dataset reflects the surface changes caused by the increase

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TABLE I
DETAILED INFORMATION ON THE FOUR DATASETS

Name	Size	Change pixel ratio	Spatial resolution	Event	Pre-event image	Post-event image
Shuguang	921×593	$\approx 1:20$	8m	Building Construction	June 2008 / Radarsat-8	Sept. 201 / Google Earth
Gloucester I	2325×4135	$\approx 1:14$	0.65m	Flooding	July 2006 / QuickBird 2	July 2007 / TerraSAR-X
Gloucester II	554×990	$\approx 1:7$	$\approx 25m$	Flooding	Sept. 1999 / SPOT	Nov. 2000 / ERS-1
California	2000×3500	$\approx 1:21$	$\approx 15m$	Flooding	Jan. 2017 / Landsat 8	Feb. 2017 / Sentinel 1-A



(a) (b) (c)

Fig. 8. Shuguang dataset. (a) Optical image. (b) SAR image. (c) Ground truth.

TABLE II
HYPERPARAMETERS FOR THE PRE-TRAINED DIFFUSION MODELS

hyperparameter	value
diffusion_steps	1000
noise_schedule	linear
learning rate	2e-5
in_channels	3
learn_sigma	True
num_heads	1
num_head_channels	64
attention_resolutions	32,16,8
schedule_sampler	uniform

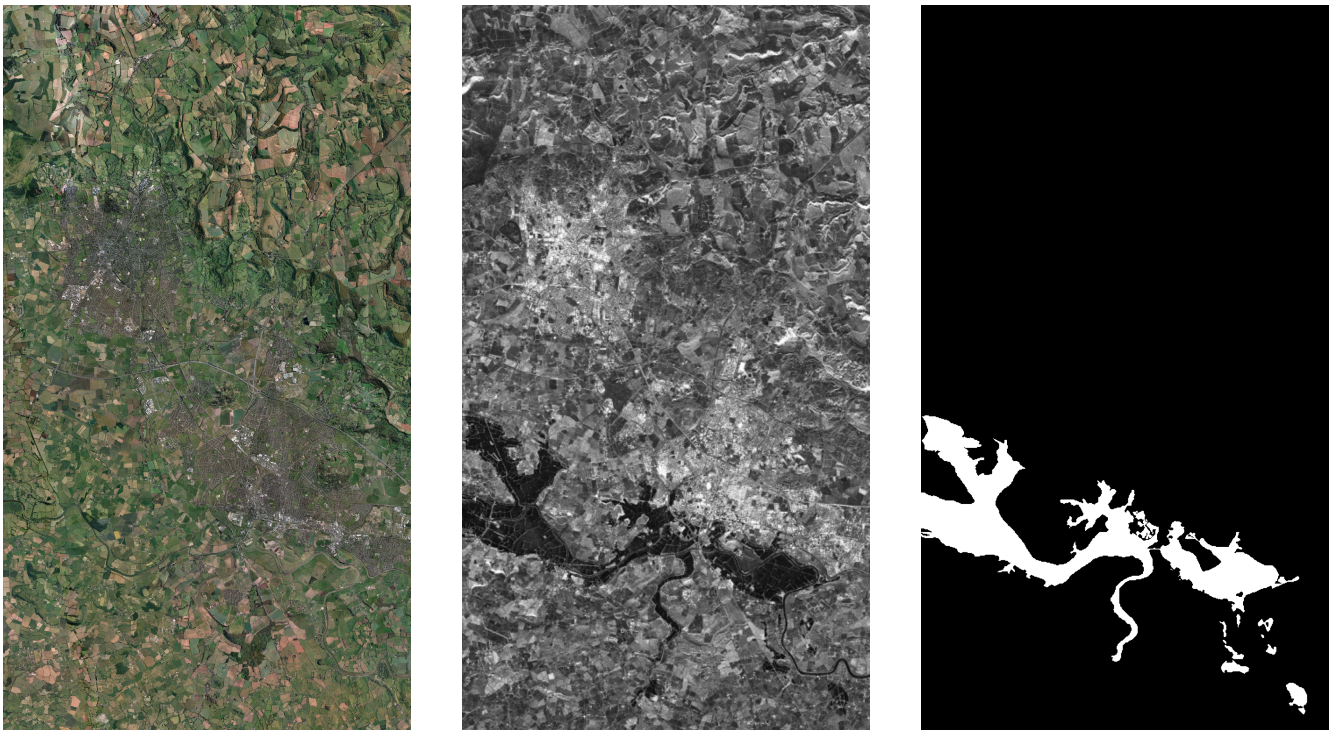
or decrease of buildings on a farmland in Shuguang, Dongying, China [44], [45]. The optical RGB image with size 921 × 593 × 3 and C-band SAR image with size 921 × 593 × 1 were obtained by Radarset-8 and Google Earth, respectively. The ground truth was manually annotated by experts based on prior knowledge.

2) *Gloucester I dataset:* As shown in Fig. 9, the second dataset shows the land cover changes caused by a flood in Gloucester, U.K. [44], [45], [46]. The optical and SAR images were acquired by Quickbird 2 in July 2006 and TerraSAR-X in July 2007, respectively, with the image size of 2325 × 4135 and the resolution of 0.65 m. The ground truth was generated manually by an expert, integrating some prior information.
3) *Gloucester II dataset:* The third dataset was again acquired at Gloucester, recording another flood event, and was used in the 2009–2010 IEEE GRSS Data Fusion Competition [46]. Fig. 10 shows the images of the dataset. The pre-event optical image was acquired by SPOT in September 1999, and the post-event SAR image was acquired by ERS-1 in November 2000. The ground truth was provided by the IEEE Data Fusion Technical Committee. Their image sizes are 554 × 990 with a spatial resolution of 25 m.
4) *California dataset:* As shown in Fig. 11, the last dataset consists of three images captured in California, United States [9], with the image size of 2000 × 3500 and the resolution of 15 m. The optical image was acquired by Landsat 8 on January 5, 2017, while the SAR image was taken by Sentinel 1-A. Unlike the above datasets, the SAR image contains three bands consisting of the VV and VH polarization bands and their ratio bands. The optical image contains 11 bands from deep blue up to thermal infrared. The ground truth was obtained by calculating the ratio and setting a threshold between two other single-polarization Sentinel-1 images at the same time.

For each dataset, we adjust images with channels other than three to three channels through band combination, and normalize the pixel values of all images to $[-1, 1]$. To fully utilize the data, each image is cropped into patches of 256 × 256 with an overlap of 50%, and random horizontal and vertical flipping and rotation are applied during training. For the optical and SAR images in each dataset, we pre-trained separate diffusion models using the same set of hyperparameters, as listed in Table II. Once trained, their parameters will be frozen and not updated during subsequent CD.

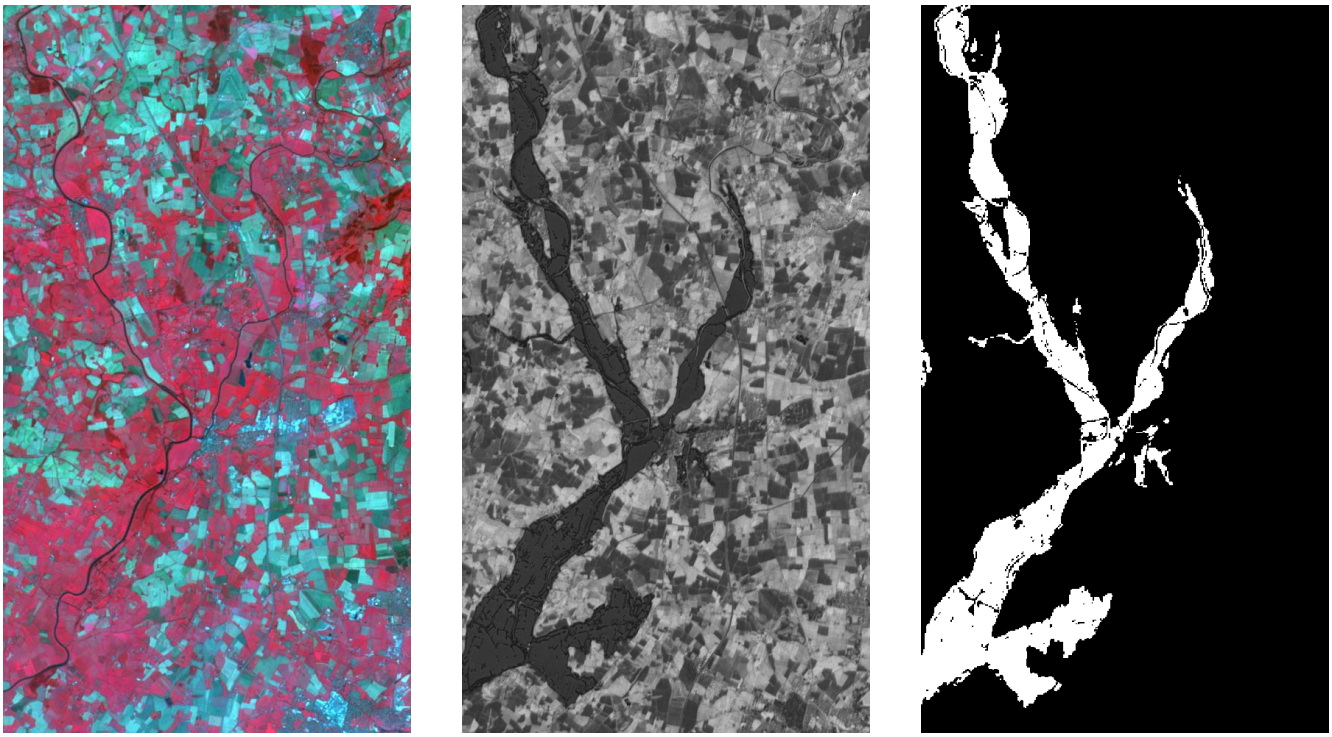
The CD experiments were conducted using the PyTorch framework on three NVIDIA RTX 3090 GPUs, with a batch size of 8 per GPU. The decoder parameters were initialized using the publicly available SNUNet [41] checkpoint pre-trained on the large-scale homogeneous CD dataset CDD, while the remaining parameters are initialized using Kaiming normalization. The learning rate was set to $1e^{-3}$ and decayed

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(a) (b) (c)

Fig. 9. Gloucester I dataset. (a) Optical image. (b) SAR image. (c) Ground truth.



(a) (b) (c)

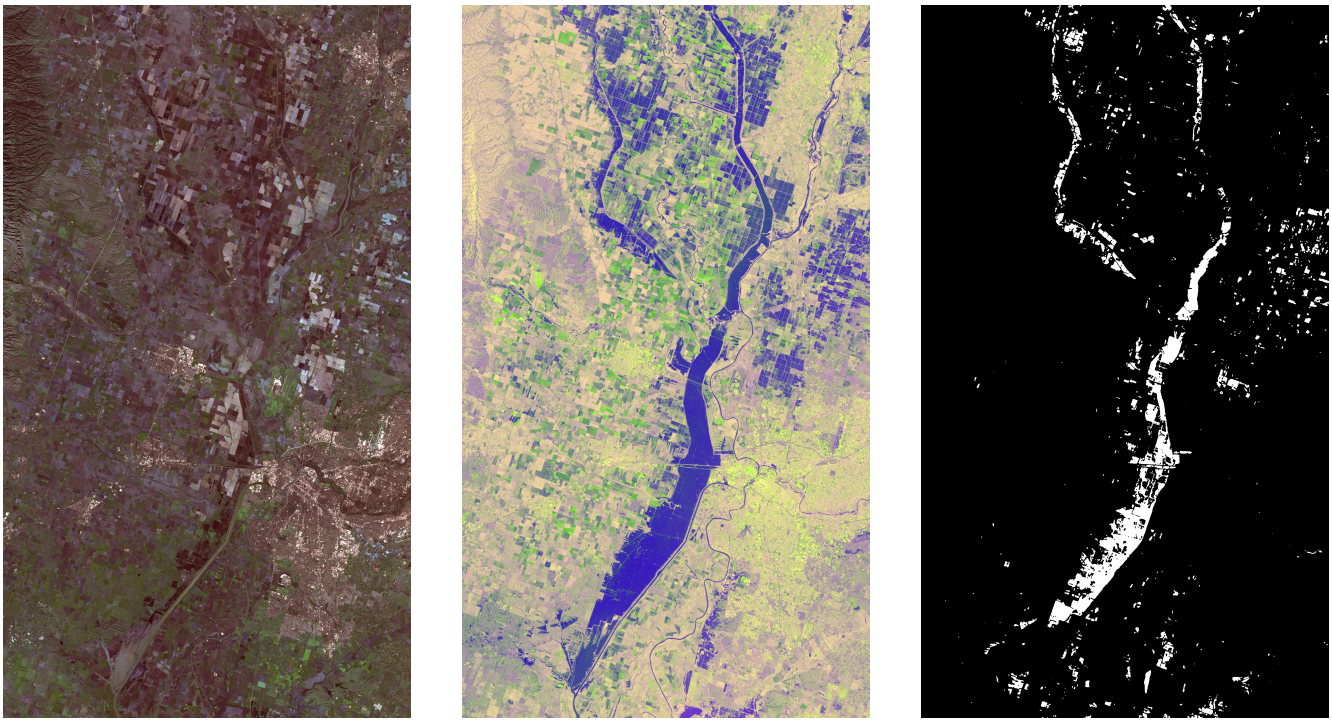
Fig. 10. Gloucester II dataset. (a) Optical image. (b) SAR image. (c) Ground truth.

by half every 70 epochs, for a total of 800 training epochs. We used the AdamW optimizer with a weight decay factor of 0.9. The final model was selected based on the epoch achieving the highest $F1$-score on the validation set.

B. Evaluation Metrics and Comparison Methods

Since the CD task can be viewed as a binary classification problem, with changing pixels representing positive classes and unchanged pixels representing negative classes,

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(a) (b) (c)

Fig. 11. California dataset. (a) Optical image. (b) SAR image. (c) Ground truth.

we evaluate the performance of each model using four frequently used metrics: precision, recall, $F1$-score, and overall accuracy (OA). These metrics can be calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{35}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{36}$$

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{37}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{38}$$

The metrics were calculated regarding the change region as positive samples. So even in the case of imbalanced categories, precision, recall, and $F1$-score can still accurately reflect the performance of each model in detecting change areas. OA is affected by category imbalance, which can be used to evaluate the overall detection performance of the model for all categories.

To verify the effectiveness of the proposed method, we compared it with eight representative multisource CD methods. Among them, cGAN, ACE Net, MTCDN, X-Net, and Codealign are representative IT-based methods, and the first three require adversarial training. The remaining SCCN, HPT, and IRG-McS belong to FT-based methods.

1) SCCN [10] is a left-right symmetric deep convolutional coupling network consisting of one convolutional layer and multiple coupling layers on each side. The two input images from both sides of the network are transformed into a common feature space to have a consistent feature representation. Then, the difference map is computed, and finally, a thresholding algorithm is applied to obtain the final CD maps.
2) cGAN [29] translates the optical image into a SAR image, then reduces the pixel differences between the translated image and the SAR image by an approximation network, and finally compares them directly to obtain the final change map.
3) HPT [7] obtains pre-difference and post-difference maps by forward and backward transformations. Then the two maps are fused, and a noise filter is applied to improve the robustness of the CD results.
4) IRG-McS [47] represents the structure of each image by constructing K-nearest neighbor graphs and then compares the graphs within the same image domain by graph mapping to compute the before and after difference images. The changes are detected by Markov cosegmentation modeling.
5) X-Net [13] first obtains the change prior by computing the affinity matrix. Then, two full convolutional networks are composed to map the multisource data from one domain to another, thereby aligning the heterogeneous features.
6) ACE-Net [13] extends the X-Net framework by introducing a latent space. A fifth network is added as a discriminator to ensure that multisource images are simultaneously consistent in terms of class distribution and feature space location.
7) Codealign [48] also utilizes a change prior. The basic idea is to align the coding layers of two self-encoders and treat them as a common latent space, so that the output of one encoder can be the input to both decoders,

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thus reconstructing the data in the original domain in the other encoder.

8) MTCDN [35] integrates a CD network into CycleGAN. It contains two generators and discriminators for optical and SAR image domains. The generators are responsible for IT, and the discriminators are used to realize image recognition and CD.

For the supervised methods, namely HPT and MTCDN, their training sets are the same as those used in Diffpurifier. The remaining self-supervised or unsupervised methods only calculate their metrics on the test set to ensure fairness in the evaluation.

C. Experiments on the Shuguang Dataset

We first present the results on the Shuguang dataset. This dataset is relatively small, and the impact of class imbalance makes cross-validation unsuitable. Given that the change areas are primarily concentrated in two regions, we use the change area in the top-left corner as the test set, with the remaining regions used for training.

The visual results of each method are shown in Fig. 12. The green and red indicate areas of change for incorrect and missed detections. The white and black represent places where changes were accurately recognized and areas where they remained constant. Obviously, our method detects the most complete regions [Fig. 12(l)], with very few red and green areas. cGAN [Fig. 12(e)] performs the worst, showing large patches of green and red regions. The detection results of X-Net [Fig. 12(h)], ACE-Net [Fig. 12(i)], and Codealign [Fig. 12(j)] using affinity matrices are quite similar, displaying clear land boundaries, indicating that their effectiveness in aligning multisource images through IT is not good. Although HPT [Fig. 12(f)] and IRG-McS [Fig. 12(g)] do not use DL, they show significantly fewer green areas, especially IRG, which only misclassifies some edges of the actual change areas. MTCDN [Fig. 12(k)] shows the detection results closest to ours, but with more missed change areas.

The quantitative evaluation results, as shown in Table III, are consistent with the qualitative analysis. It can be seen that our proposed Diffpurifier performs excellently in recall, $F1$-score, and OA, with only a slightly lower precision compared to MTCDN.

TABLE III
QUANTITATIVE EVALUATION RESULTS ON THE SHUGUANG DATASET

Method	Precision	Recall	F1	OA
SCCN [10]	51.42	67.50	58.37	78.23
cGAN [29]	46.56	44.80	45.66	75.89
HPT [7]	77.35	**75.63**	76.48	89.48
IRG-McS [47]	81.59	75.44	78.39	90.60
X-Net [13]	63.93	60.70	62.28	83.37
ACE-Net [13]	67.22	63.91	65.53	84.79
Codealign [48]	66.61	62.77	64.63	84.46
MTCDN [35]	**93.02**	74.61	**82.81**	**92.99**
Ours	**92.34**	**91.96**	**92.16**	**96.46**

D. Experiments on the Gloucester I Dataset

The Gloucester I is the largest dataset in the experiment, and except for MTCDN and IRG-McS, none of the other comparative methods have been tested on this dataset in their original papers, so hyperparameters need to be manually selected.

For MTCDN and IRG-McS, we used the hyperparameters provided in the original papers. SCCN is a DL-based FT-based method, and due to the absence of translation constraints, the loss function is relatively simple, consisting of only two terms. The method performed best when the coefficient λ, which controls the probability map weight, was set to 0.04. HPT is an ML-based FT-based method, and we used the reimplementation code from the GitHub repository.[1] The process of obtaining the difference image does not require hyperparameter tuning. During the clustering process, we manually set the threshold to choose the optimal result. For the remaining IT-based methods, which either require adversarial training or involve many hyperparameters, we have tried various adjustments, but the results were still unsatisfactory.

According to Fig. 13, cGAN [Fig. 13(e)] performs the worst, and HPT [Fig. 13(f)] has many false detection areas (green). Among the methods that use affinity matrices to obtain change priors, Codealign [Fig. 13(j)] outperforms ACE-Net [Fig. 13(i)] and X-Net [Fig. 13(h)] slightly. Although SCCN [Fig. 13(d)] and IRG-McS [Fig. 13(g)] have fewer false detection areas, their detected regions are incomplete, particularly along land boundaries where many discontinuities exist. The detection performance of MTCDN [Fig. 13(k)] is quite similar to our method [Fig. 13(h)], with both high homogeneity in detected regions and better alignment with land boundaries.

Quantitative evaluation in Table IV shows that our method significantly outperforms the comparative methods in terms of precision. Although recall is slightly lower than MTCDN, our $F1$-score still leads, demonstrating that Diffpurifier performs optimally in detecting change areas. Since OA includes the model's precision in detecting unchanged areas, if the model tends to predict unchanged regions, OA will appear inflated, even if its ability to detect changed regions is poor. This is the reason why most methods have an OA exceeding 90%, yet the $F1$-score remains poor.

E. Experiments on the Gloucester II Dataset

The Gloucester II dataset is relatively small and exhibits a relatively more balanced proportion of pixels in the changing and unchanged areas, so we applied cross-validation to this dataset. Specifically, we divided the dataset into six equal parts, using one part for testing and the remaining five parts for training in each iteration.

In Fig. 14, we present the detection results of the comparative methods from a single experiment. It is evident that most

[1] https:github.com/llu025/Heterogeneous_CD

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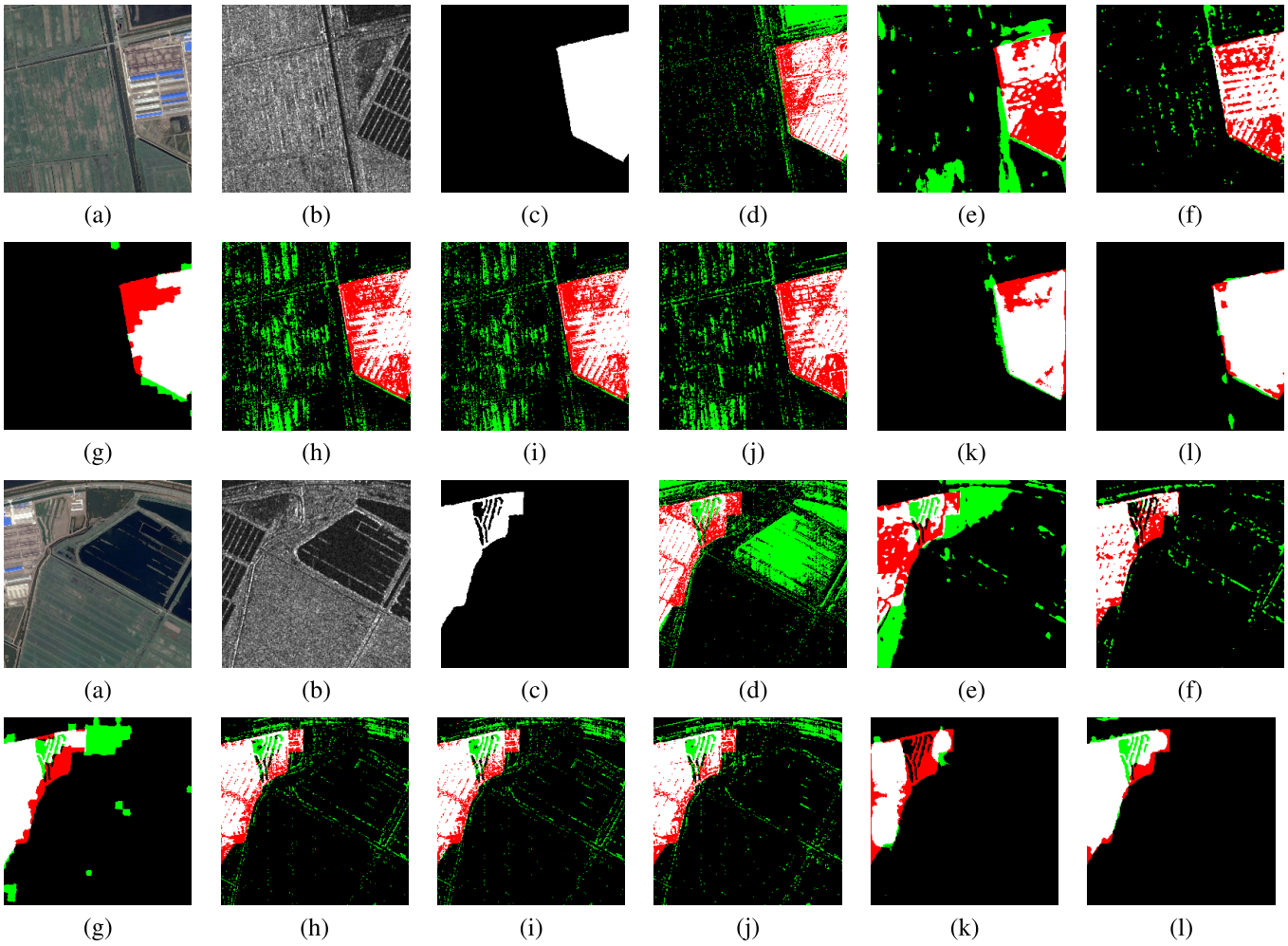


Fig. 12. Visualization results on the Shuguang dataset. (a) Optical image. (b) SAR image. (c) Ground truth. (d) SCCN. (e) cGAN. (f) HPT. (g) IRG_McS. (h) X-Net. (i) ACE-Net. (j) Codealign. (k) MTCDN. (l) Ours.

TABLE IV
QUANTITATIVE EVALUATION RESULTS ON THE GLOUCESTER I DATASET

Method	Precision	Recall	F1	OA
SCCN [10]	56.52	57.34	57.07	97.07
cGAN [29]	17.75	33.37	23.13	81.44
HPT [7]	27.29	72.25	39.61	85.87
IRG-McS [47]	49.04	60.63	54.22	95.31
X-Net [13]	19.99	36.30	25.78	90.43
ACE-Net [13]	22.12	42.30	29.05	90.53
Codealign [48]	29.59	71.49	41.86	90.90
MTCDN [35]	**67.04**	**77.87**	**72.05**	**97.23**
Ours	**81.55**	**75.32**	**78.32**	**98.08**

TABLE V
QUANTITATIVE EVALUATION RESULTS ON THE GLOUCESTER II DATASET

Method	Precision	Recall	F1	OA
SCCN [10]	67.23	74.83	70.79	93.21
cGAN [29]	45.57	42.67	43.46	86.32
HPT [7]	52.01	66.74	58.15	88.80
IRG-McS [47]	76.85	80.28	78.23	94.78
X-Net [13]	58.75	77.45	66.56	91.21
ACE-Net [13]	53.38	77.74	62.76	89.33
Codealign [48]	56.72	53.75	57.85	89.66
MTCDN [35]	**86.31**	**93.58**	**89.69**	**97.36**
Ours	**90.04**	**95.90**	**92.87**	**98.23**

methods have large patch-like misdetection areas. Although IRG-McS [Fig. 13(g)] and MTCDN [Fig. 13(k)] show relatively complete detection regions, both methods incorrectly classify the unchanged crossroad areas located within the change zones as changes. In contrast, our method not only achieves high homogeneity in the detection regions but also performs well in detecting linear roads.

The average value of the sixfold cross-validation metrics is used as the metric for each algorithm, as shown in Table V. Relatively low spatial resolution helps detect large-scale flood disasters in this dataset, and most algorithms exhibit relatively high recall. However, in terms of precision, only IRG-McS and MTCDN perform slightly better. Our method outperforms all other methods across all metrics.

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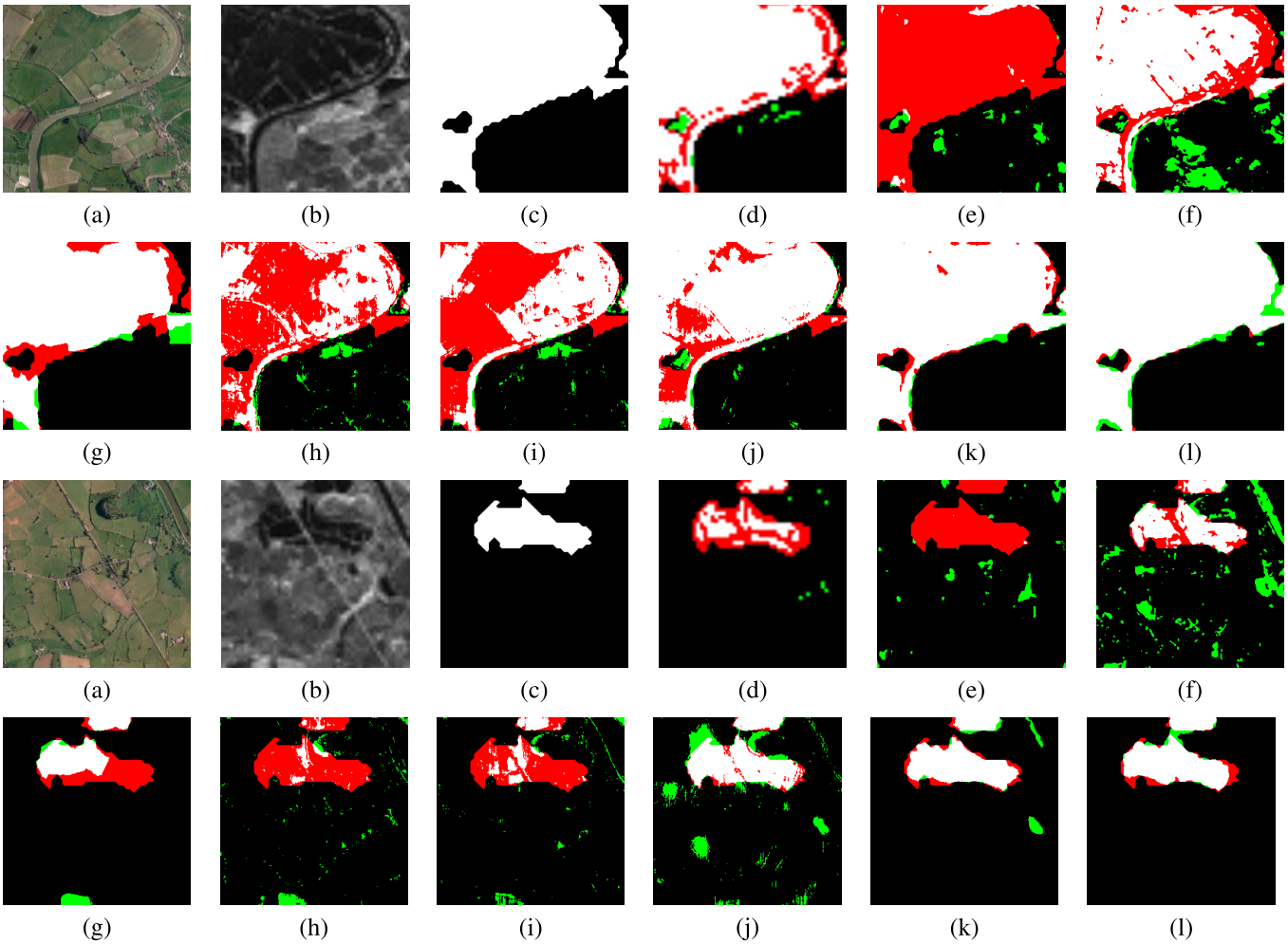


Fig. 13. Visualization results on the Gloucester I dataset. (a) Optical image. (b) SAR image. (c) Ground truth. (d) SCCN. (e) cGAN. (f) HPT. (g) IRG_McS. (h) X-Net. (i) ACE-Net. (j) Codealign. (k) MTCDN. (l) Ours.

F. Experiments on the California Dataset

The California dataset is relatively challenging, as it not only has the lowest proportion of change pixels but also features highly scattered change areas. Based on Fig. 15, most algorithms exhibit large areas of misdetections, represented by significant green regions. Compared with single-polarization data, the multipolarization SAR images in the California dataset have more complex features and noise. SCCN [Fig. 13(d)] and HPT [Fig. 13(f)] clearly transmit more noise to the resulting images. However, overall, there is no significant difference between IT-based methods and FT-based methods, and both perform poorly. Table VI demonstrates that most algorithms achieve high recall, yet their precision is relatively low. Although our method does not achieve the highest precision or recall, it still performs the best in terms of the $F1$-score, indicating that our method strikes a good balance between conservativeness and the scope of detection. While MTCDN has higher precision and slightly better $F1$-score, visual results [Fig. 13(k)] suggest that it tends to classify unchanged areas within the change regions as changes.

TABLE VI
QUANTITATIVE EVALUATION RESULTS ON THE CALIFORNIA DATASET

Method	Precision	Recall	F1	OA
SCCN [10]	33.46	77.15	46.68	89.09
cGAN [29]	39.86	35.90	37.78	92.68
HPT [7]	38.54	85.19	53.07	90.67
IRG-McS [47]	49.99	81.53	61.98	93.81
X-Net [13]	42.62	**86.36**	57.07	91.96
ACE-Net [13]	42.05	83.80	56.00	91.85
Codealign [48]	44.15	**87.71**	58.74	92.37
MTCDN [35]	**70.16**	67.99	**69.06**	**96.22**
Ours	**69.47**	71.58	**70.51**	**96.29**

V. DISCUSSION

In this section, we explore the impact of different hyperparameters on detection results, analyze the computational burden of Diffpurifier, and demonstrate the flexibility of the model through an additional multimodel experiment.

A. Model Hyperparameters Analysis

1) Hyperparameters of Pre-Trained Diffusion Models: The translation algorithm DDIBs [39] is key to simplifying the loss function of Diffpurifier and eliminating translation constraints.

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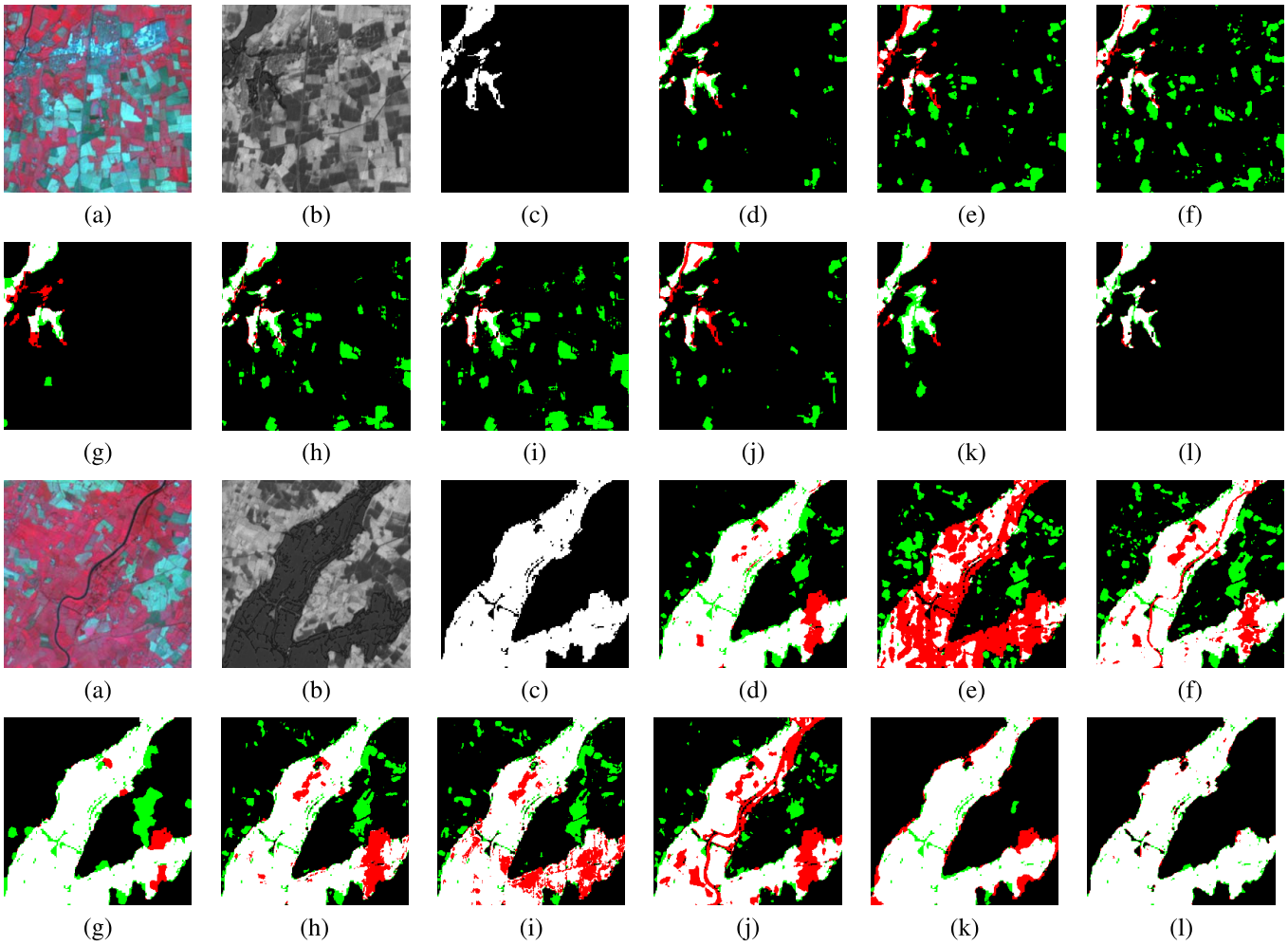


Fig. 14. Visualization results on the Gloucester II dataset. (a) Optical image. (b) SAR image. (c) Ground truth. (d) SCCN. (e) cGAN. (f) HPT. (g) IRG_McS. (h) X-Net. (i) ACE-Net. (j) Codealign. (k) MTCDN. (l) Ours.

Although the original paper has been tested on synthetic and traditional datasets, its performance on remote sensing datasets remains unknown. Therefore, we need to verify whether DDIBs adhere to cycle consistency on multisource remote sensing images and explore the optimal general approach for translating optical images into SAR images.

First, we conduct cycle consistency tests on the individual domains of optical and SAR images, where images are mapped to the latent space using the forward ODE of the diffusion model from the source domain, and then mapped back to the source domain for reconstruction using the reverse ODE. In this test, we primarily adjust hyperparameters related to the diffusion process to observe the image reconstruction results.

Fig. 16 shows a small part of the test results for single-domain cycle consistency under different parameters. By reconstructing optical images, we found that for images with a size of 256×256, adding noise linearly increases the result better than adding cosine. By learning to add noise variance, clearer images can be restored. However, using exponential moving averages for model parameter optimization did not achieve better results, and increasing the number of sampling steps may lead to color distortion in the image.

Next, by adjusting hyperparameters related to the UNet network structure of the diffusion model, we seek a general approach for IT across multiple source domains. Fig. 17 shows the translation results of optical to SAR images under different adjustments. Based on the consideration of image style and the degree of image feature preservation, we choose to convert into YUV space and set UNet's base channel to 256. Intuitively, the resolution of different attention applications makes the network focus on features at different levels. The more residual blocks there are, the more fine-grained content one can learn, but the less sensitive to coarse-grained features.

Since our task is CD, we do not aim for highly realistic translation effects but instead strive to capture land cover representations best. The final hyperparameters of the pre-trained diffusion models, as shown in Table II, represent the general approach for IT that we have developed for downstream tasks. This setting can be applied to different datasets and sensors.

2) Impact of Different Combinations of Timesteps: According to the previous description, the magnitude of timesteps t can reflect the noise level contained in the latent variable, which is the depth of purification of the diffusion process. In the combination $t = t_1, t_2, \ldots, t_n$, n represents the number of purification times. Table VII shows how the evaluation

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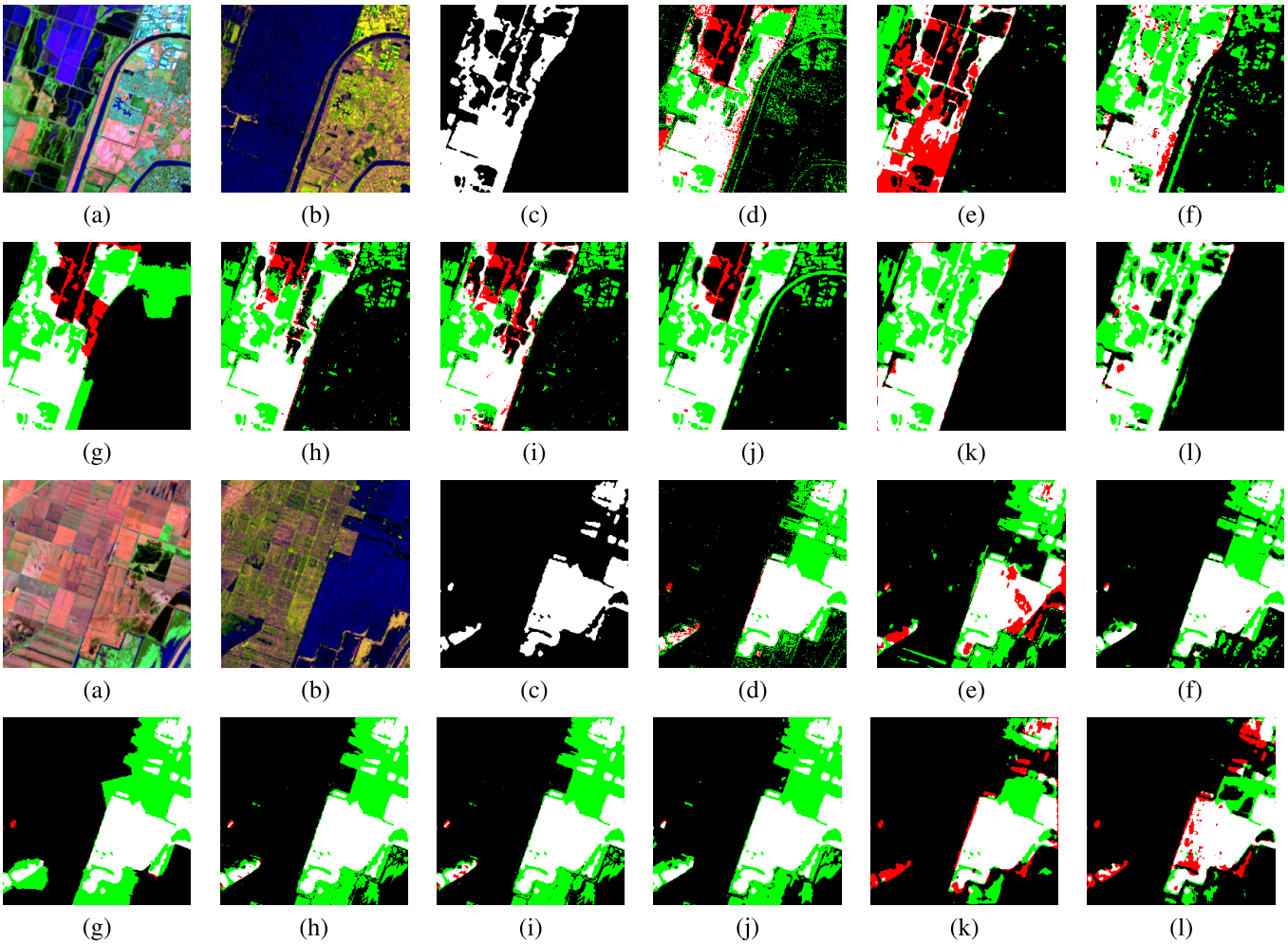


Fig. 15. Visualization results on the California dataset. (a) Optical image. (b) SAR image. (c) Ground truth. (d) SCCN. (e) cGAN. (f) HPT. (g) IRG_McS. (h) X-Net. (i) ACE-Net. (j) Codealign. (k) MTCDN. (l) Ours.

TABLE VII
RESULTS OF DIFFERENT t COMBINATIONS ON THE GLOUCESTER II DATASET

Timestep	Precision	Recall	F1	OA
1	92.30	97.84	94.99	98.35
2	92.96	97.83	95.02	98.47
3	91.75	**98.49**	95.02	98.35
4	91.62	**98.57**	94.97	98.33
5	91.75	98.46	94.99	98.34
2,3	93.31	97.50	95.36	98.49
3,5	92.13	98.26	95.10	98.38
4,5	**93.63**	97.53	**95.54**	**98.55**
1,2,3	93.15	97.66	95.35	98.48
1,4,5	93.54	97.53	**95.49**	**98.53**
2,3,5	**93.64**	97.33	95.45	98.52

metrics of CD vary with different combinations of t. We use a DDIM [38] with a stride of 200 for the initial 1000 timesteps inference. A smaller stride is also feasible, but it will significantly prolong the experimental time.

According to Table VII, we can find that the $F1$-score is higher when t is set to 2 or 3 for a single purification. This confirms that the ability of UNet to resist noise and extract semantic information from images is not directly proportional but rather checks and balances each other. Moreover, compared to a single purification, more purification times can achieve more stable $F1$-score and OA, which is likely due to the increased credibility of multiscale features and enhanced suppression of noise.

Overall, shallow single purification can still achieve good results. This also validates the powerful ability of diffusion models to extract inherent semantic information from images. In practical applications, from the perspective of time and efficiency, a smaller t can be chosen; from the perspective of stability detecting, we recommend selecting the timestep combination with the best $F1$-score performance.

3) Hyperparameters of CD Network: Diffpurifier uses superpixels combined with an attention mechanism to improve the homogeneity of the detection result image and reduce potential noise. To verify its effectiveness, we removed this module and directly set the final output channel of pixel-level change features to 2 to obtain the final detection result.

Table VIII shows the quantitative ablation experiment results. The improvement of OA and $F1$-score indicates

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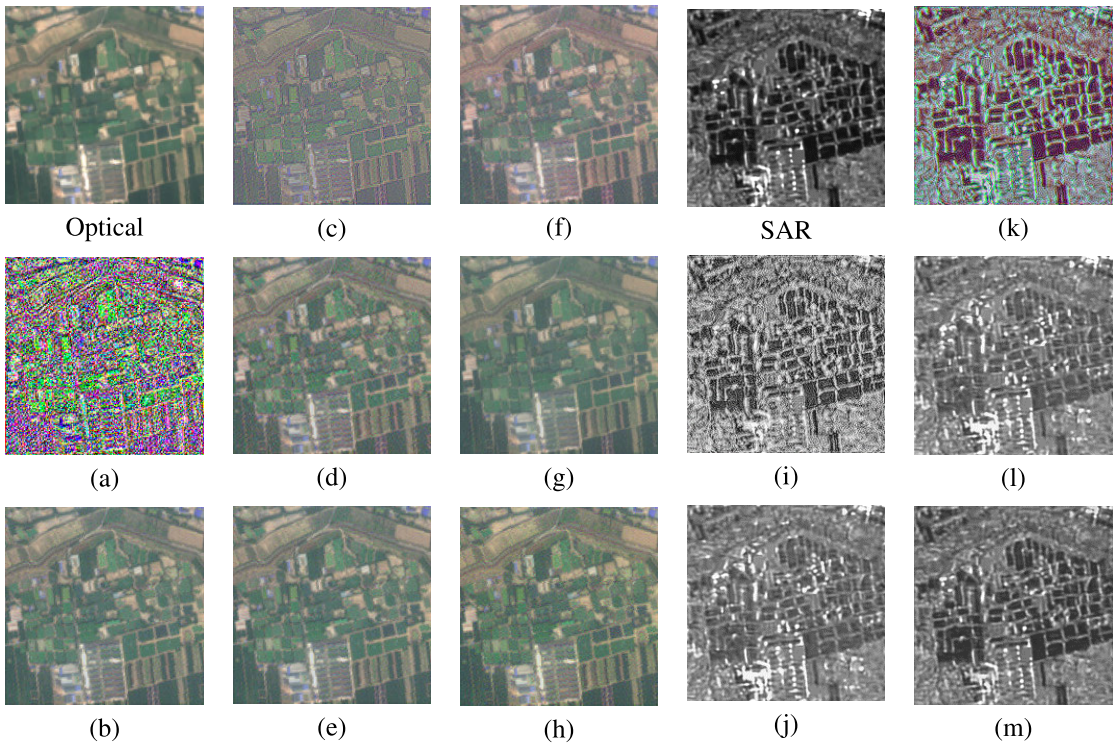


Fig. 16. Cycle consistency test results for single domain under different parameters. For the optical domain, (a) with cosine noise schedule, (b) with linear noise schedule, (c) use exponential moving average (EMA) model, (d) with learned the variance of noise, (e) without learning the variance of noise, (f) 6000 training steps, (g) 4000 training steps, and (h) 1000 training steps. For the SAR domain, (i) and (j) correspond to the maximum value of the noise schedule of 0.03 and 0.02, respectively. (k)–(m) correspond to 8000, 6000, and 4000 training steps, respectively.

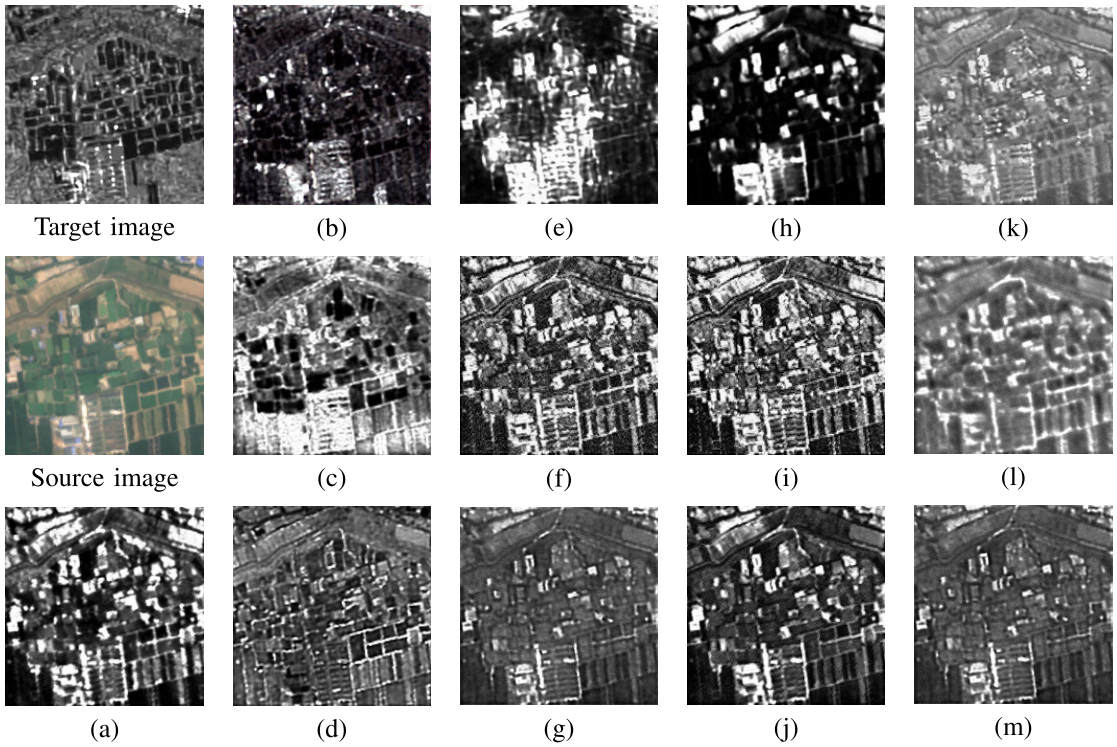


Fig. 17. Translation results under different scenarios. (a)–(d) Denote the results of first transforming the optical image into a pan image, YCbCr, YUV, and Lab space before translation, respectively. (e)–(g) Denote the results when the base channel of UNet is 16, 32, and 256, respectively. (h)–(j) Denote the results when UNet's attention is 1, 4, and 64, respectively. (k)–(m) Denote the results when the number of blocks of UNet is 4, 3, and 2, respectively.

that the addition of superpixels and attention mechanisms enhances the comprehensive detection performance of the model. To further investigate the influence of superpixel size on detection results, we changed M from 64 to 729 with a fixed SLIC iteration of 5 to find the most suitable number of superpixels.

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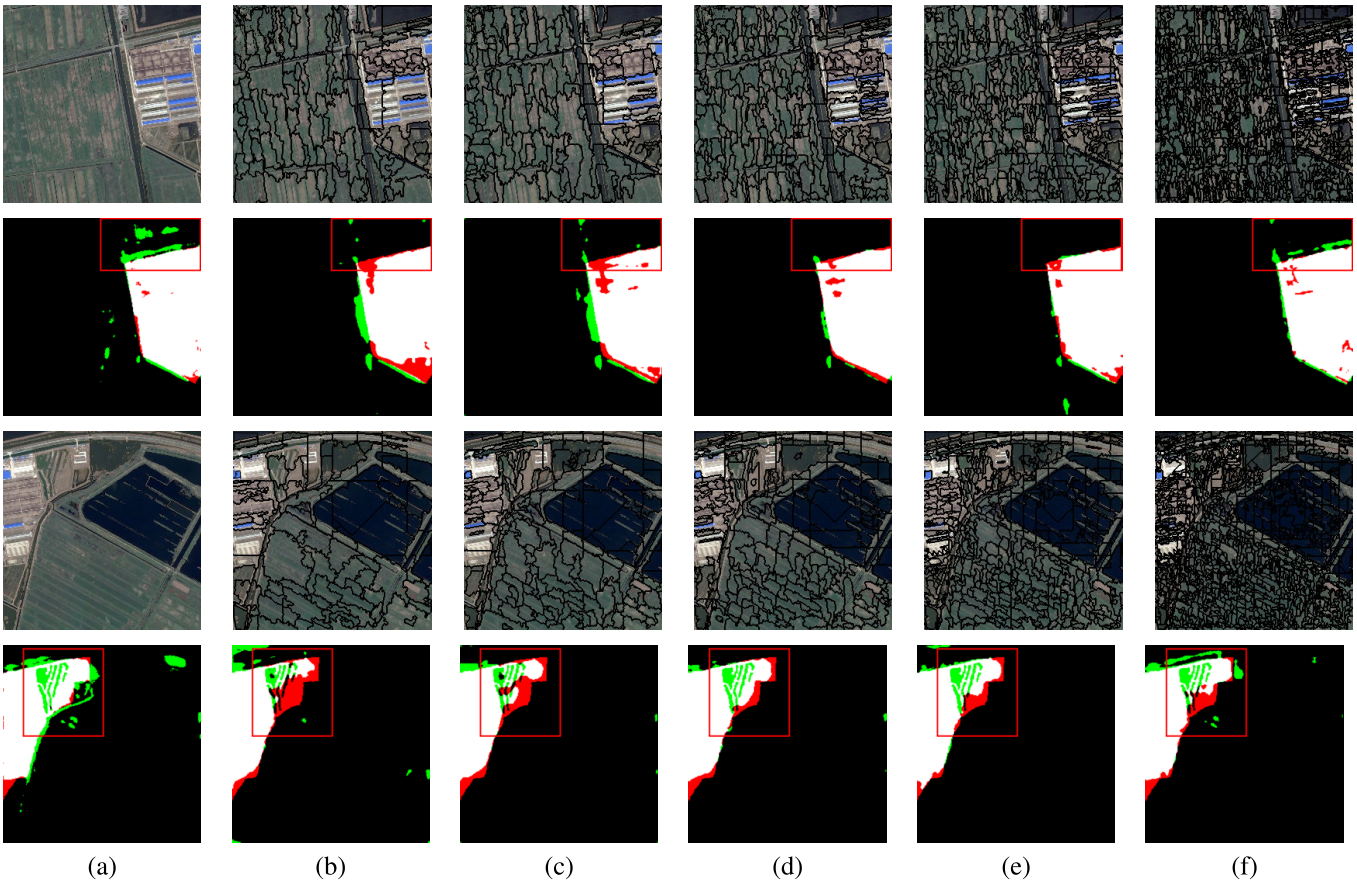


Fig. 18. Segmentation boundaries and corresponding CD results overlaid on the original RGB image for different values of M. (a)–(f) Correspond to the cases without superpixels and with superpixel counts of 64, 100, 144, 256, and 576, respectively.

TABLE VIII
ABLATION EXPERIMENTS FOR SUPERPIXEL AND ATTENTION MECHANISMS ON THE SHUGUANG DATASET

Methods	Precision	Recall	F1	OA
w/o superpixel	84.66	**96.91**	90.37	95.33
Diffpurifier	**92.34**	91.96	**92.16**	**96.46**

TABLE IX
ABLATION EXPERIMENTS FOR OPTIMAL NUMBER OF SUPERPIXELS ON THE SHUGUANG DATASET

M	Precision	Recall	F1	OA
64	86.63	84.83	85.72	94.08
100	89.42	86.89	88.13	94.71
144	**93.35**	90.90	**92.11**	**96.47**
256	**92.34**	**91.96**	**92.16**	**96.46**
576	88.62	**94.23**	91.34	95.95
729	91.40	87.76	89.55	95.36

Fig. 18 shows the segmentation boundaries and corresponding CD results overlaid on the original RGB image for different values of M. As shown in rows 1 and 2, within the range of 64–256, an increase in the number of superpixels leads to a reduction in misclassified areas (green regions), thereby improving precision. However, as shown in rows 3 and 4, overly loose segmentation causes superpixels to inadequately represent neighborhood features, resulting in the omission of subtle changes, which increases the red regions and reduces recall.

Table IX quantitatively reflects the impact of the number of superpixels on detection. When M exceeds 256, the $F1$-score begins to decline. Combining this with the last column of the figure, this may be due to an excessive number of superpixels, which causes the homogeneous regions within the segmented blocks to fail in conveying complete semantic information. This not only undermines the smoothing effect but also increases computational complexity. Therefore, we ultimately set M to 256.

Regarding the remaining hyperparameters of the CD network, we refer to SNUNet [41] to set the learning rate, the optimizer, and the number of channels for decoders, and the number of iterations J for superpixels to be the same as in ESCNet [42]. These parameters were previously summarized as the optimal solution, so we will not discuss them further.

B. Model Complexity Analysis

To further evaluate the overall performance of Diffpurifier, we compared the computational efficiency of different algorithms. Table X summarizes the trainable parameters and

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(a) (b) (c)

Fig. 19. Gloucester I dataset. (a) Pre-event image. (b) Post-event image. (c) Ground truth.

TABLE X
TRAINABLE PARAMETERS AND INFERENCE TIME OF DL-BASED COMPARISON METHODS

Method	Number of Parameters	Inference Time(s)
SCCN [10]	5374	0.52 (CPU)
cGAN [29]	265151	5.11 (CPU)
ACE-Net [13]	436527	61.1 (CPU)
X-Net [13]	123474	21.44 (CPU)
Codealign [48]	108420	11.22 (CPU)
MTCDN [35]	48.06M	0.54 (GPU)
Ours	8.08M (+552.81M)	3.21 (GPU)

inference time of DL-based comparison methods when running an image with a size of 256 × 256.

Except for MTCDN, the other comparison methods are based on tensorflow1.x and do not match our GPU version, so we only calculated their CPU inference time. While the parameter sizes of these methods are smaller, they struggle to handle larger images. For example, on the California dataset, these methods require resizing images by a factor of 4 to make them runnable. And their detection accuracy is also limited, particularly when transferred to different datasets, as they require frequent adjustments of hyperparameters. However, it must be acknowledged that our method still has considerable room for improvement in inference time. For example, SCCN, even when run on CPU, is much faster than our method.

Moreover, although in our method, the 8.08M parameters correspond to the trainable part of the CD network is smaller than the MTCDN (also based on PyTorch and using GPU), and the 552.81M parameters correspond to the pre-trained model does not require further training in practice, it still affects the batch size, leading to lower GPU utilization. Additionally, our inference time is much slower than that of MTCDN, so balancing the representational power, model size, and inference speed of the model will be the key direction for improvement in the future.

TABLE XI
QUANTITATIVE EVALUATION RESULTS ON THE TOULOUSE DATASET

Method	Precision	Recall	F1	OA
EENPPI [49] *	51.53	59.89	55.39	85.38
Codealign [48] *	71.25	41.06	52.09	86.90
MTCDN [35]	63.37	**85.73**	72.87	92.93
Ours	**89.35**	64.01	**74.59**	**95.17**

*Results for EENPPI and Codealign were derived from their respective original papers.

C. Model Generalization Analysis

Diffipurifier has achieved good results between optical and SAR sensors, and to validate its portability to other types of data, we tested it on a multisource dataset provided by ESA Earth Observation Portal.[2] As shown in Fig. 19, this dataset is from Toulouse, France and reflects the changes caused by building construction. It consists of a pair of optical images from different spectral bands, with a resolution of 0.52 m and a cloud cover of 0%. The first optical image was captured by the Pleiades sensor in May 2012, and the second optical image is a screenshot from Google Earth, captured by the WorldView2 satellite (with three spectral bands: red, green, and blue) on July 11, 2013 [49]. The WorldView2 VHR image was coregistered in [50] to match the Pleiades image.

We cut each image into 196 small patches of size 256 × 256, overlapping by 50%. About 138 patches are used for training, and 58 patches are used for testing. We applied the same parameter settings used in the other datasets and compared the results with several other multisource CD algorithms. The results are summarized in Table XI.

From Table XI, it can be seen that our method outperforms the compared algorithms in terms of precision, $F1$-score, and OA. Specifically, although the algorithm has a slightly lower recall, it achieved the highest precision (89.35%), indicating that it effectively detected true positives and minimized false positives to the greatest extent possible. Our $F1$-score

[2]https://earth.esa.int/web/eoportal/satellite-missions

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(74.59%) also demonstrates a good balance between precision and recall, further highlighting the robustness of our method in detecting changed regions.

This small experiment shows that our algorithm not only performs well in optical and SAR datasets but also can be effectively extended to data with different sensors. No matter how obvious the differences between sensors are, our algorithm can still lead to $F1$-score without changing the hyperparameters, demonstrating strong portability and generalization ability.

VI. Conclusion

This article presents Diffpurifier, a method for CD between optical and SAR images. It utilizes pre-trained diffusion models and ODEs to translate optical images into SAR images and incorporates superpixel-pixel associations in the subsequent CD process to increase the homogeneity of the CD map. Compared to earlier IT-based methods, our approach combines IT and feature extraction, allowing the model to achieve high detection accuracy without requiring additional translation constraints during training. Experimental results on four public datasets show that, in terms of $F1$-score, Diffpurifier outperforms the second-best method by an average of approximately 5%. Without adjusting hyperparameters, our method also performs well when migrating to data from other multisource sensors, demonstrating stable generalization ability.

Despite the excellent performance of Diffpurifier, its main limitation lies in the large number of parameters in the pre-trained models, leading to relatively longer inference times. Therefore, future work will focus on balancing model size, inference time, and detection capability. Additionally, to fully leverage the generative capability of diffusion models, we plan to explore the use of conditional constraints to directly generate CD maps, which would both increase inference speed and further enhance detection performance.

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